

IT Security 2

Unit 8 - Controlling Application Usage with App-ID

Instructor: Brian Krause

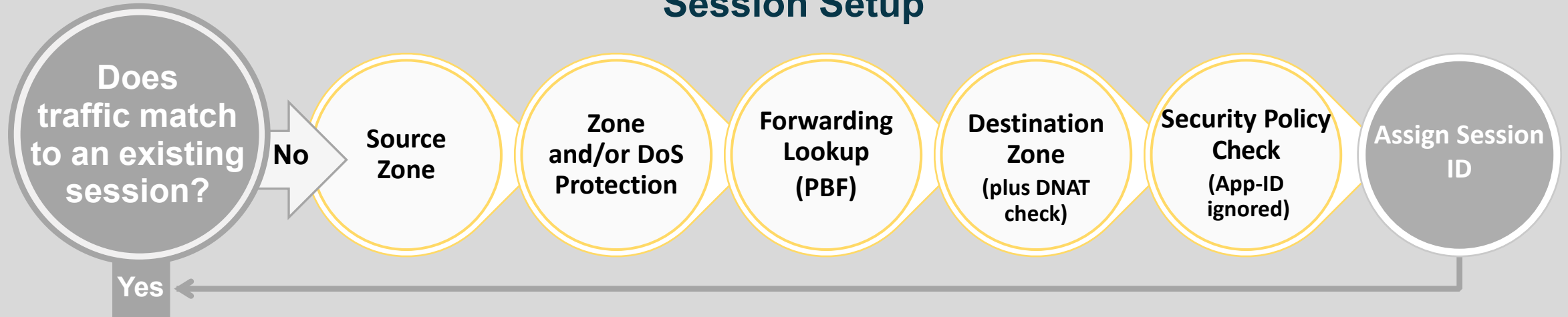
Announcements

Controlling Application Usage with App-ID

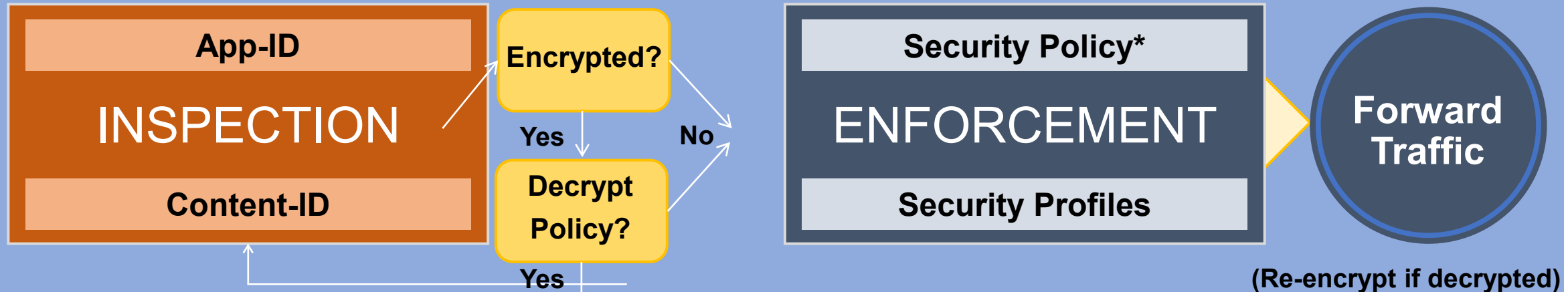
Some slides were adapted from Palo-Alto for use in this class. Content is a mesh between vendor provided information and instructor comments.

Flow Logic of the Next-Generation Firewall

Session Setup



Inspection and Enforcement



* Policy check relies on pre-NAT IP addresses

CONTROLLING APPLICATION USAGE WITH APP-ID



PROTECTION IS NOT A PRINCIPLE BUT AN EXPEDIENT

- App-ID reduces the attack surface
- App-ID concepts and operation
- Configure App-ID objects
- Unknown and encrypted application traffic
- Migrating to an App-ID-based Security Policy
- Updating App-ID

Learning Objectives

After you complete this module, you should be able to:

- Identify how App-ID reduces the attack surface
- Describe App-ID concepts and operation
- Configure App-ID-based policy rules
- Update App-ID application database





App-ID reduces the attack surface

App-ID concepts and operation

Configure App-ID objects

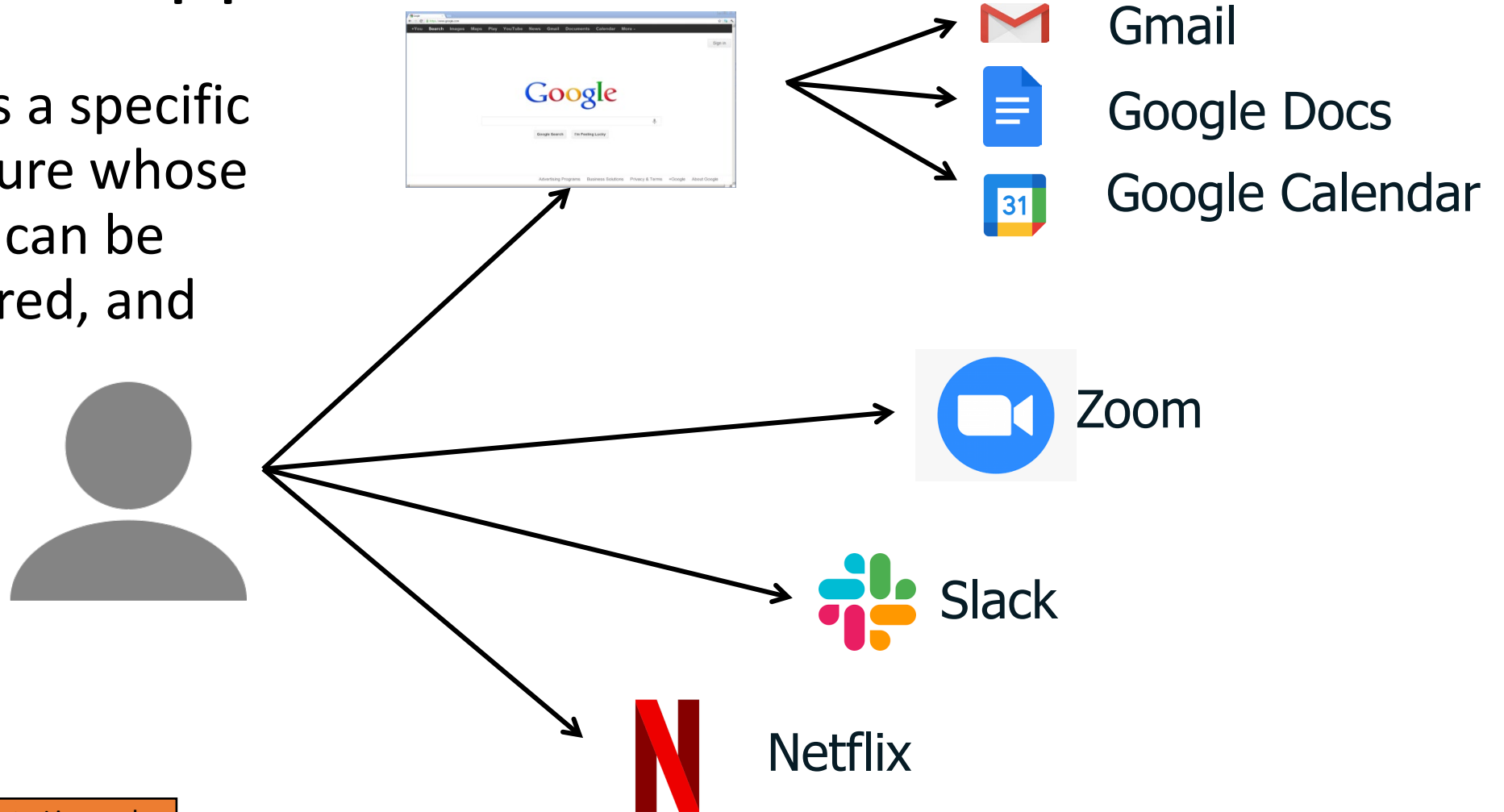
Unknown and encrypted application traffic

Migrating to an App-ID-based Security Policy

Updating App-ID

What Is an Application?

An *application* is a specific program or feature whose communication can be labeled, monitored, and controlled.



For a Tech Doc about this topic, log into Live and search for "App-ID Overview."

What Is App-ID?

Objects > Applications

For a more information about Applications, use
applipedia.paloaltonetworks.com

Search

All

Clear Filters

3615 matching applications

CATEGORY ^	SUBCATEGORY ^	RISK ^	TAGS ^	CHARACTERISTIC ^
1437 business-systems	54 audio-streaming	1813 1	29 eLearning	37 Data Breaches
451 collaboration	24 auth-service	984 2	87 Enterprise VoIP	640 Evasive
374 general-internet	41 database	565 3	31 Entertainment Video	667 Excessive Bandwidth
323 media	2 design	361 4	18 G Suite	49 FEDRAMP
494 networking	89 email	144 5	28 Palo Alto Networks	1 FINRA
786 saas	73 encrypted-tunnel			111 HIPAA
2 unknown	45 erp-crm			86 IP Based Restrictions
	272 file-sharing			100 New App ID

NAME	CATEGORY	SUBCATEGORY	TECHNOLOGY	RISK	TAGS	STANDARD PORTS
☐ alipay	collaboration	social-business	client-server	2	Web App	tcp/80,443
☐ aporeto	business-systems	general-business	client-server	1		tcp/dynamic
☐ avid-nexis	business-systems	storage-backup	client-server	2		tcp/7238-7245
☐ aws-iot	networking	infrastructure	client-server	1		tcp/dynamic
☐ azure-govt-cloud-storage	general-internet	file-sharing	browser-based	1	Web App	tcp/443
☐ bacnet						

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Add

Delete

Clone

Enable

Disable

Import

Export

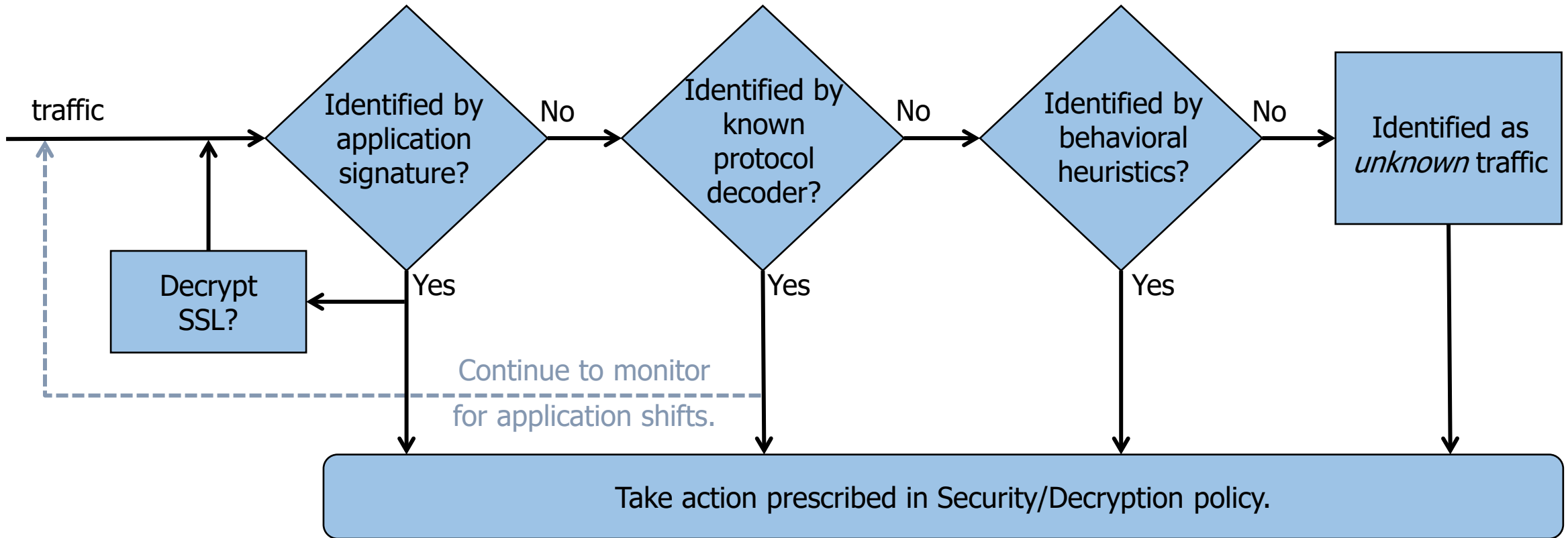
PDF/CSV

Review Policies

Edit Tags

App-ID Application Identification

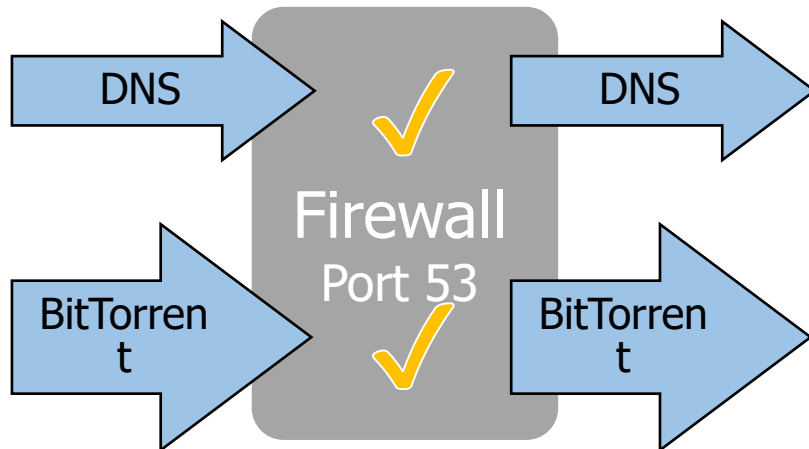
App-ID identifies applications in traffic observed by the firewall.



Port-Based Versus Next-Generation Firewalls

Traditional Firewalls

Firewall Rule: **ALLOW Port 53**



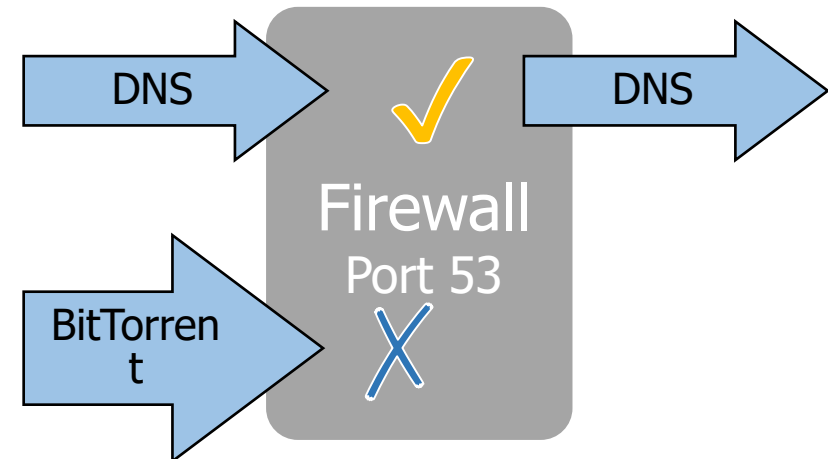
Packet on port 53: **Allow**

Packet on port 53: **Allow**

Visibility: Port 53 allowed

Palo Alto Networks Firewalls with App-ID

Firewall Rule: **ALLOW DNS**



DNS = DNS: **Allow**

BitTorrent $\neq$ DNS: **Deny**

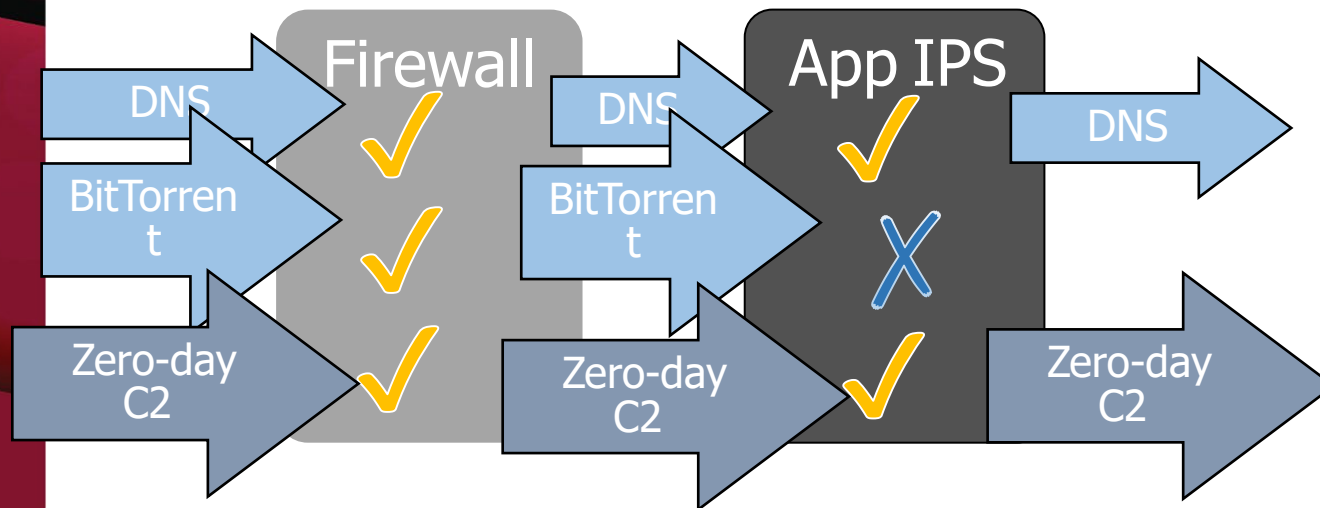
Visibility: BitTorrent detected and blocked

Zero-Day Malware: IPS Versus App-ID

Legacy Firewalls

Firewall Rule: **ALLOW Port 53**

Application IPS Rule: **Block BitTorrent**



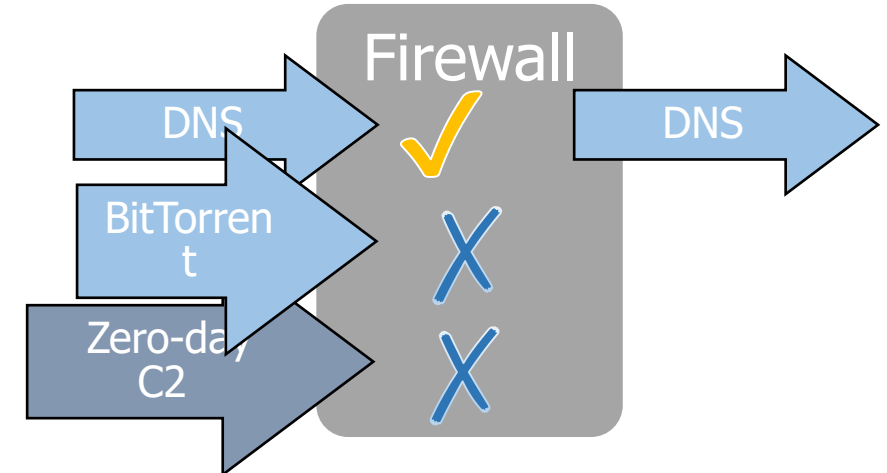
Packet on port 53: **Allow**

C2 ≠ BitTorrent: **Allow**

Visibility: Packet on port 53 allowed

Palo Alto Networks Firewalls with App-ID

Firewall Rule: **ALLOW DNS**

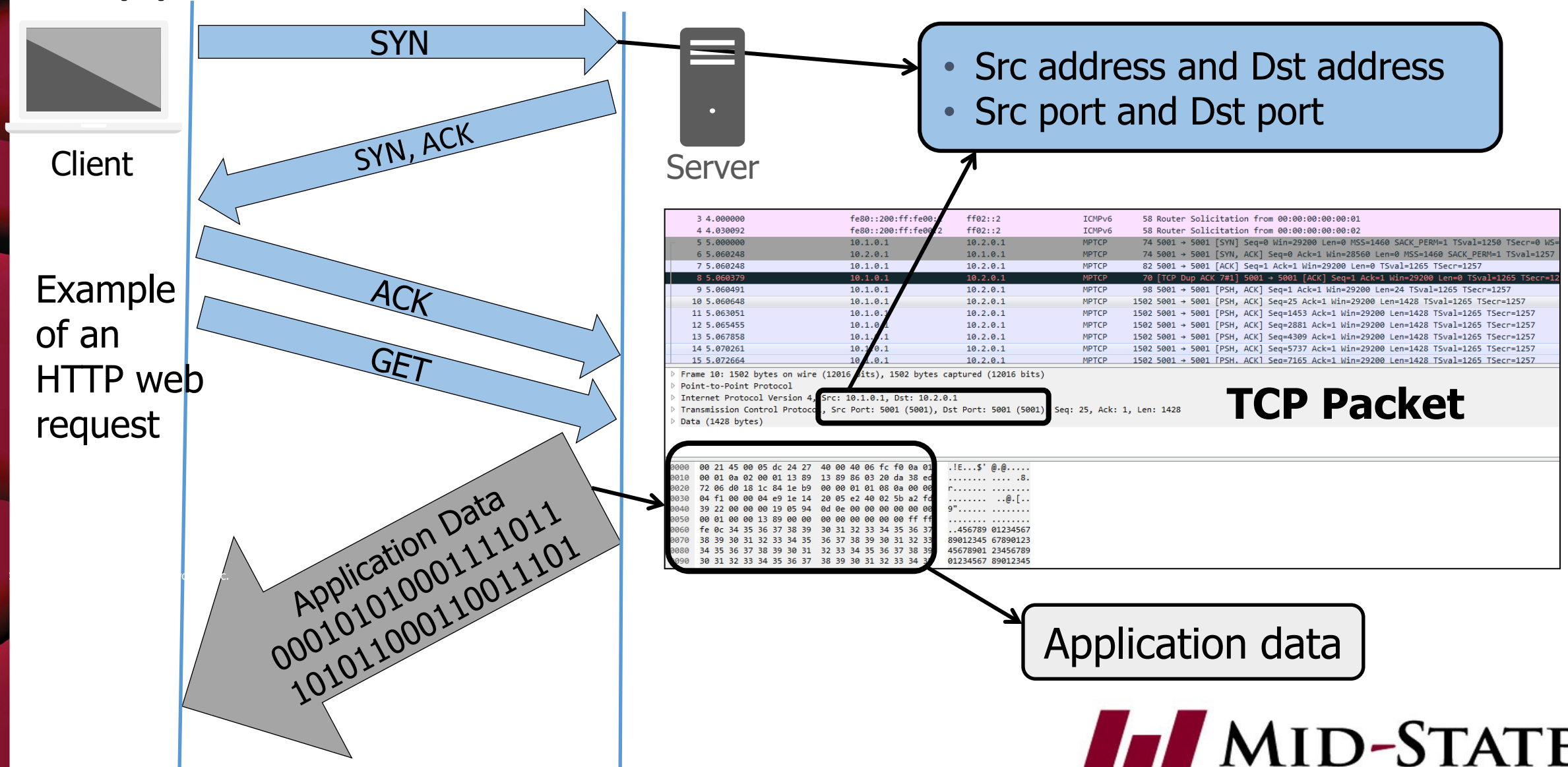


DNS = DNS: **Allow**

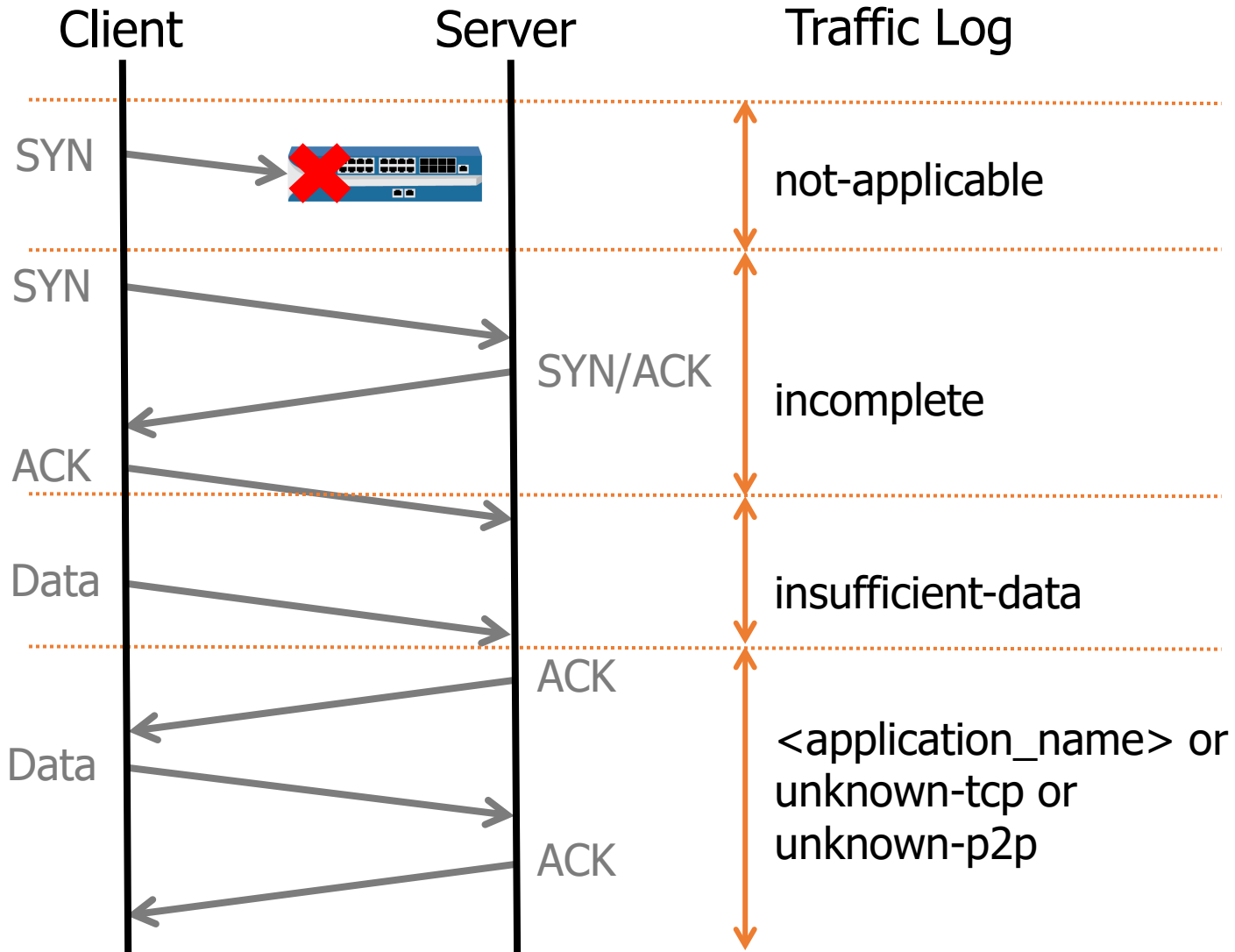
C2 ≠ DNS: **Deny**

Visibility: Unknown traffic
detected and blocked

App-ID and TCP



Classifying (Labeling) TCP Traffic



- not-applicable:
 - Traffic dropped per policy before application identified.
- incomplete:
 - Three-way handshake did not complete or was followed by no data.
- insufficient-data:
 - Not enough payload for identification.
- unknown-tcp:
 - Unidentifiable TCP traffic.
- unknown-p2p:
 - Unidentifiable peer-to-peer traffic.

App-ID and UDP

Lightweight UDP Packet

```

> Ethernet II, Src: 3ComCorp_c7:87:49 (00:04:75:c7:87:49), Dst: 3ComCorp_dd:bb:3a (00:04:75:dd:bb:3a)
> Internet Protocol Version 4, Src: 139.133.204.176, Dst: 139.133.204.183
> Lightweight User Datagram Protocol, Src Port: 32768 (32768), Dst Port: 1234 (1234)
> Data (12 bytes)

```

```

0000  00 04 76 dd bb 3a 00 04 75 c7 87 49 08 00 45 00  ..v....u..I..E.
0010  00 28 1a 6a 40 00 40 88 6f 71 8b 85 cc b0 8b 85  .(.j@.@.oq.....
0020  cc b7 80 00 04 d2 00 00 38 45 68 65 6c 6c 6f 20  .....8Ehello
0030  77 6f 72 6c 64 0a 00 00 00 00 00 00  world... ....

```

```

> Internet Protocol Version 4, Src: 139.133.204.176, Dst: 139.133.204.183
> Lightweight User Datagram Protocol, Src Port: 32768 (32768), Dst Port: 1234 (1234)
> Data (12 bytes)

```

```

0000  00 04 76 dd bb 3a 00 04 75 c7 87 49 08 00 45 00  ..v....u..I..E.
0010  00 28 1a 6a 40 00 40 88 6f 71 8b 85 cc b0 8b 85  .(.j@.@.oq.....
0020  cc b7 80 00 04 d2 00 00 38 45 68 65 6c 6c 6f 20  .....8Ehello
0030  77 6f 72 6c 64 0a 00 00 00 00 00 00  world... ....

```

```

> Internet Protocol Version 4, Src: 139.133.204.176, Dst: 139.133.204.183
> Lightweight User Datagram Protocol, Src Port: 32768 (32768), Dst Port: 1234 (1234)
  Data (12 bytes)
    Data: 68656c6c6f20776f726c640a
    [Length: 12]

```

```

0000  00 04 76 dd bb 3a 00 04 75 c7 87 49 08 00 45 00  ..v....u..I..E.
0010  00 28 1a 6a 40 00 40 88 6f 71 8b 85 cc b0 8b 85  .(.j@.@.oq.....
0020  cc b7 80 00 04 d2 00 00 38 45 68 65 6c 6c 6f 20  .....8Ehello
0030  77 6f 72 6c 64 0a 00 00 00 00 00 00  world... ....

```

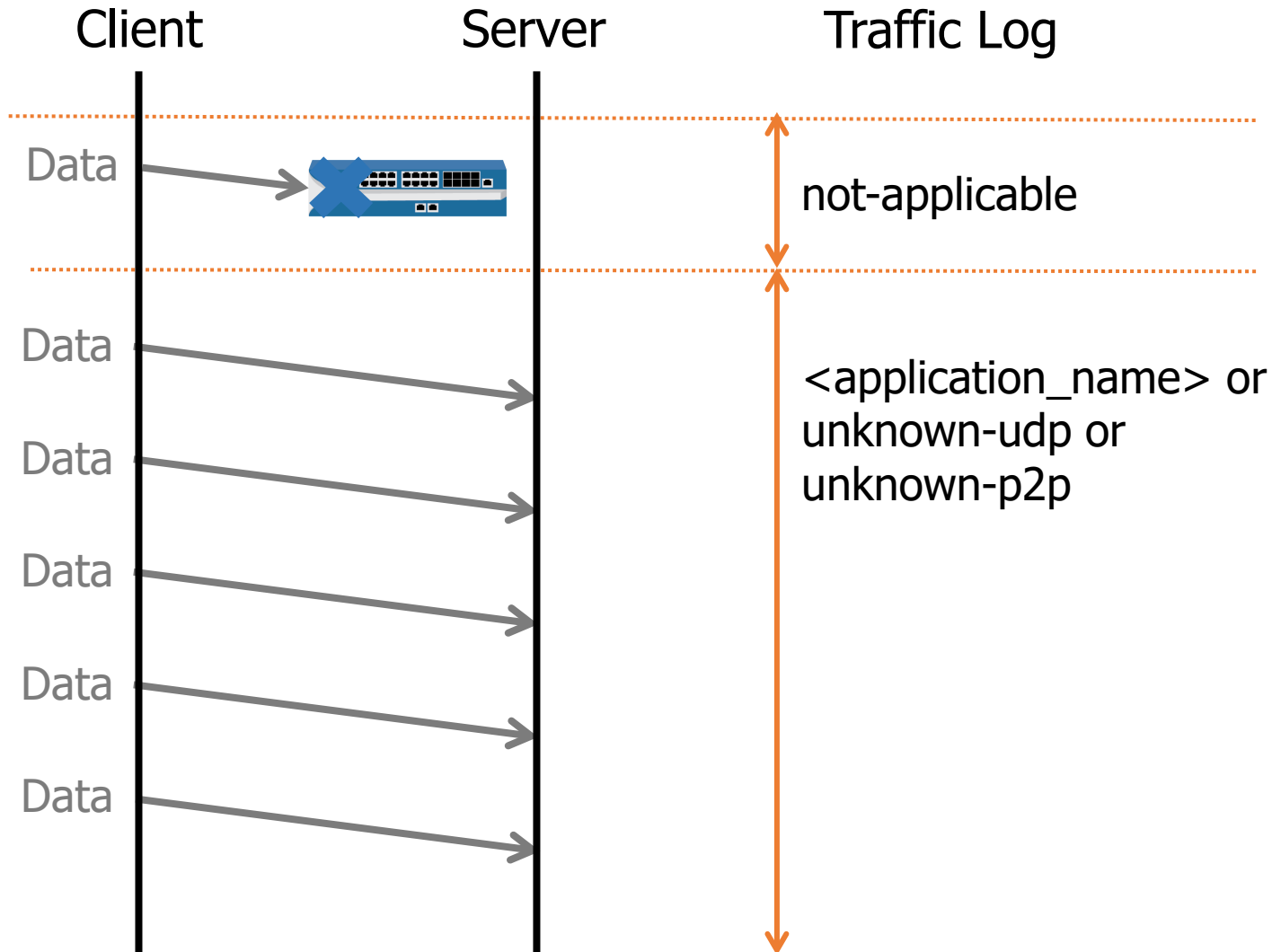
The first UDP packet

Src address and Dst address

Src port and Dst port

Application data

Classifying (Labeling) UDP Traffic



- not-applicable:
 - Traffic dropped per policy before application identified
- unknown-udp:
 - Unidentifiable UDP traffic
- unknown-p2p:
 - Unidentifiable peer-to-peer traffic

App-ID reduces the attack surface



App-ID concepts and operation

Configure App-ID objects

Unknown and encrypted application traffic

Migrating to an App-ID-based Security Policy

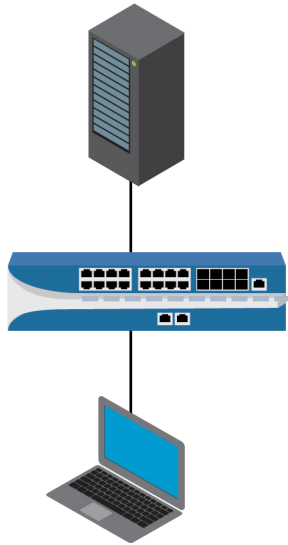
Updating App-ID

Application Shifts

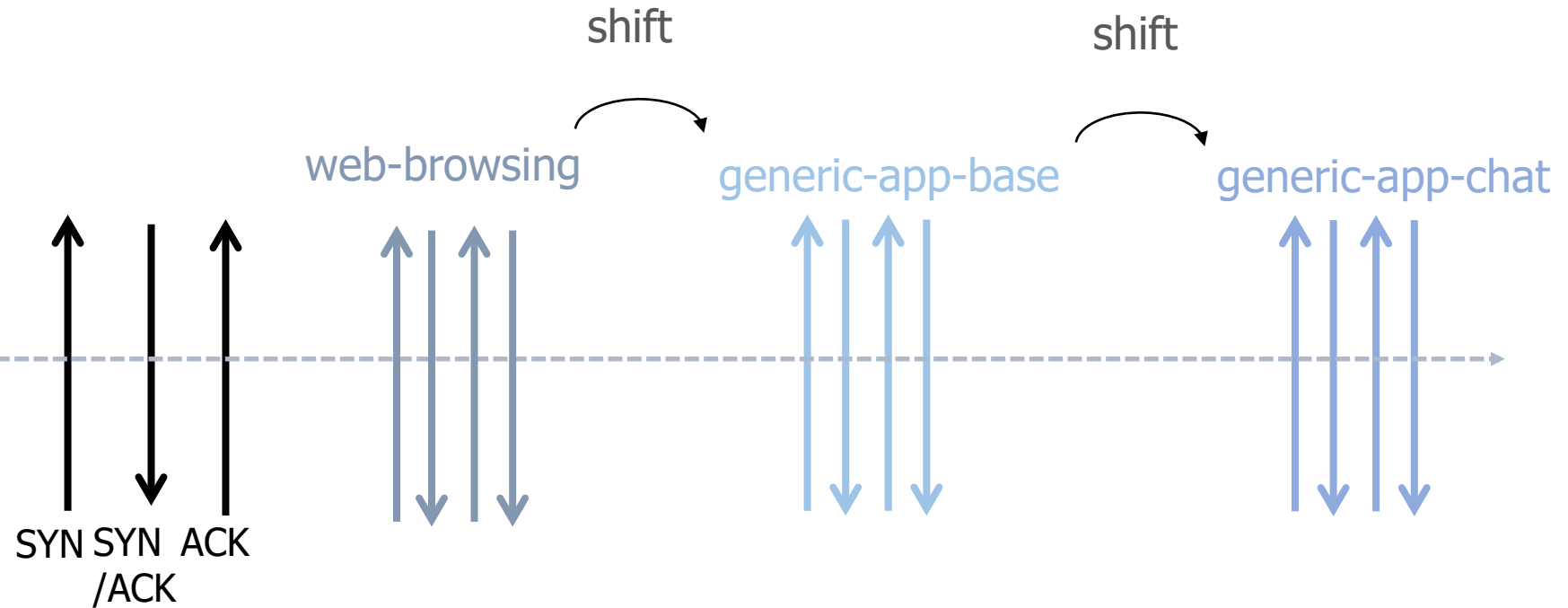
Network traffic can shift from one application to another during a session.

www.generic-app.com

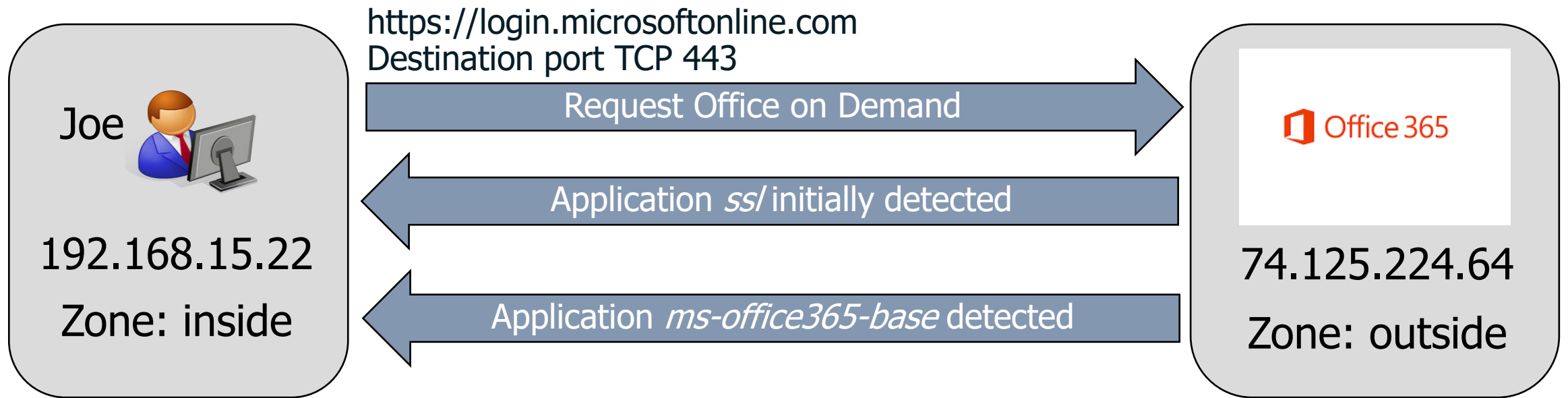
Web Server



Web Client



Application Dependencies



	NAME	TAGS	TYPE	Source			Destination		APPLICATION	SERVICE	URL CATEGORY	ACTION
				ZONE	ADDRESS	USER	ZONE	ADDRESS				
1	Required-Apps	Users_Net	universal	Users_Net	any	any	Internet	any	ssl web-browsing	application-default	any	Allow
2	Allow-Office-365	Users_Net	universal	Users_Net	any	any	Internet	any	ms-office365-base office-on-demand sharepoint-online	application-default	any	Allow

Application shift

Note: office-on-demand is dependent on ms-office365-base and sharepoint-online

View Application Dependencies Before Modifying a Rule

Objects > Applications

The screenshot shows the Palo Alto Networks management console interface. On the left, a table lists applications under the 'Applications' category. The search bar at the top is set to 'ms-office365-base'. The application 'ms-office365-base' is selected, and its details are shown in a modal window on the right. The modal window displays the application's name, standard ports, and dependencies. The 'Depends on' field is highlighted with a red box, showing 'ssl, web-browsing'. The 'Description' field provides a detailed overview of Office 365. The 'Characteristics' section lists various attributes of the application, and the 'Options' section lists configurable settings.

Search	ms-office365-base	All	Clear Filters	
CATEGORY ^	SUBCATEGORY ^	RISK ^	TAGS ^	CHARACTERISTIC ^
9 business-systems	1 email			1 Data Breaches
3 collaboration	1 er			
1 networking	1 ge			
	1 in			
	7 of			
	1 re			
	1 so			

NAME
☐ microsoft-dynamics-crm
☐ ms-office365 (7 out of 19 shown)
☐ ms-lync-online
☐ ms-office365-base

Application

Name: ms-office365-base

Standard Ports: tcp/80,443

Depends on: ssl, web-browsing

Implicitly Uses:

Deny Action: drop-reset

Additional Information: [Office 365](#) [Wikipedia](#) [Google](#) [Yahoo!](#)

Characteristics

Evasive:	no	Tunnels Other Applications:	no
Excessive Bandwidth Use:	no	Prone to Misuse:	no
Used by Malware:	no	Widely Used:	yes
Capable of File Transfer:	no	SaaS:	yes
Has Known Vulnerabilities:	yes		

Options

Session Timeout (seconds):	30	Customize...
TCP Timeout (seconds):	3600	Customize...
TCP Half Closed (seconds):	120	Customize...
TCP Time Wait (seconds):	15	Customize...
App-ID Enabled:	yes	

Dependent applications require you to add a Security policy rule.

View Unresolved Dependencies Reported

	NAME	Source				Destination			APPLICATION	SERVICE	ACTION
		ZONE	ADDRESS	USER	DEVICE	ZONE	ADDRESS	DEVICE			
1	Users_to_Internet	Users_Net	any	any	any	Internet	any	any	ms-office365-base	application-default	Allow

Commit Status

Operation Commit

Status Completed

Result Successful

Details Configuration committed successfully

Commit **App Dependency**

1 item

RULE	COUNT
Users_to_Internet	2

1 item

APP	DETAIL
ms-office365-base	- ms-office365-base requires ssl to be allowed. - ms-office365-base requires web-browsing to be allowed.

Missing application dependencies

- A commit determines if application dependencies in any rule are satisfied by any rule.
- Unresolved dependencies are reported per rule.
- Click rule's **Count** number to view unresolved dependencies.
- Click **<rule_name>** to open and edit rule.

Implicit Applications

- Many common applications implicitly allow parent applications.
- *No need* to explicitly define these applications in a Security policy rule:
 - They are automatically allowed.

	NAME	TAGS	TYPE	Source			Destination		APPLICATION	SERVICE	URL CATEGORY	ACTION
				ZONE	ADDRESS	USER	ZONE	ADDRESS				
1	Allow-Facebook	Users_Net	universal	Users_Net	any	any	Internet	any	facebook-apps facebook-base facebook-chat	application-default	any	Allow

facebook-base implicitly allows web-browsing and ssl, so no need to add a rule for them.

For a Tech Doc about this topic, log into Live and search for "Applications with Implicit Support."

Determine Implicitly Used Applications

Objects > Applications

Search X Q All Clear Filters

CATEGORY ^	SUBCATEGORY ^	RISK ^	TAGS ^	CHARACTERISTIC ^
13 collaboration	1 email	3 1	15 Web App	3 Evasive
1 general-internet				
3 media				

NAME

- ☐ facebook
- ☐ facebook-apps
- ☐ facebook-base
- ☐ facebook-chat

Application

Name: facebook-base

Standard Ports: tcp/80,443, udp/443

Depends on:

Implicitly Uses: ssl, web-browsing

Deny Action: drop-reset

Additional Information: [Wikipedia](#) [Google](#) [Yahoo!](#)

Characteristics

Evasive:	no	Tunnels Other Applications:	yes
Excessive Bandwidth Use:	no	Prone to Misuse:	no
Used by Malware:	yes	Widely Used:	yes
Capable of File Transfer:	yes		
Has Known Vulnerabilities:	yes		

Classification

Category: collaboration

Description:

Facebook (branded as "facebook") is a social networking website launched on February 4, 2004. The free-access website is privately owned and operated by Facebook, Inc. Users can join networks organized by city, workplace, school, and region to connect and interact with other people. People can also add friends and send them messages, and update their personal profile to notify friends about themselves. The website's name refers to the paper facebooks depicting members of a campus community that some American colleges and preparatory schools give to incoming students, faculty, and

Options

Session Timeout (seconds):	30	Customize...
TCP Timeout (seconds):	3600	Customize...
UDP Timeout (seconds):	30	Customize...
TCP Half Closed (seconds):	120	Customize...
TCP Time Wait (seconds):	15	Customize...
App-ID Enabled:	yes	

App-ID reduces the attack surface

App-ID concepts and operation



Configure App-ID objects

Unknown and encrypted application traffic

Migrating to an App-ID-based Security Policy

Updating App-ID

Application Groups

Objects > Application Groups > Add

Application Group ?

Name

3 items → ×

☐	APPLICATIONS
☐	twitter
☐	facebook
☐	youtube

 Browse  Add  Delete

- Static, administrator-defined sets of applications
- Used to simplify Security and QoS policy rulebases

Application Filters

Objects > Application Filter > Add

Application Filter

NAME ☐ Apply to New App-IDs only 34 matching applications

CATEGORY ^	SUBCATEGORY ^	RISK ^	TAGS ^	CHARACTERISTIC ^
34 business-systems	207 general-business	15 1	0 App-ID Cloud Engine	7 Evasive
	783 ics-protocols	13 2	0 eLearning	3 FEDRAMP
	221 management	3 3	1 Enterprise VoIP	3 HIPAA
	69 medical	3 4	0 Entertainment Video	3 IP Based Restrictions
	34 office-programs		0 Entertainment Video	11 No Certifications
	7 software-development		0 G Suite	3 PCI
	36 software-update			2 Prone to Misuse
	46 storage-backup			

NAME	CATEGORY	SUBCATEGORY	RISK	TAGS	STANDARD PORTS	EXCLUDE
airtable						☒
airtable-base	business-system:	office-programs	1	Web App	443,80,tcp	☒
airtable-downloading	business-system:	office-programs	2	Web App	443,80,tcp	☒
airtable-editing	business-system:	office-programs	1	Web App	443,80,tcp	☒
airtable-posting	business-system:	office-programs	1	Web App	443,80,tcp	☒

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Showing Technology Column

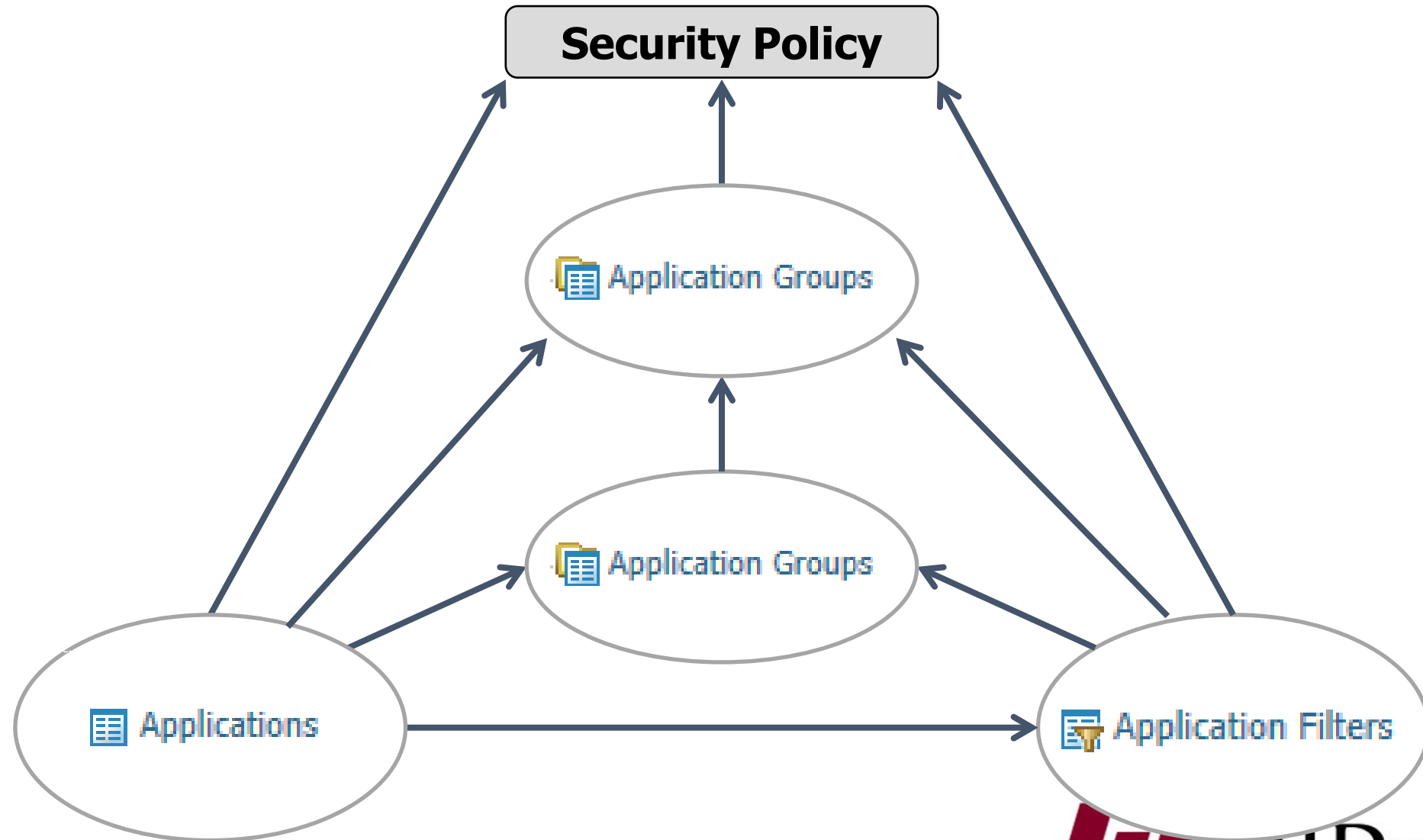
OK Cancel

Displaying 1 - 38 of 38

- Dynamic group of applications
- Created by selecting filters in the App-ID database
- Used to simplify Security, QoS, and PBF policy rulebases

Application tags can be used as application filters.

Nested Application Groups and Filters



Predefined and Custom Application Tags

Objects > Applications

Application

Name: adobe-connectnow-base Description: Adobe ConnectNow is a web meeting service that enables users to have online conferences and live collaboration using their web browser and the Adobe Flash Player. ConnectNow provides text and audio chat, video conferencing, whiteboard, text notes, screen-sharing and collaborative remote control of any of the participant's computers with their permission. It is a part of the Acrobat.com suite of cloud-based applications provided by Adobe.

Standard Ports: tcp/80,443,1935

Depends on: flash, rtmp, rtmp, ssl, web-browsing

Implicitly Uses:

Deny Action: drop-reset

Additional Information: [Wikipedia](#) [Acrobat.com](#) [Google](#) [Yahoo!](#)

Characteristics

Evasive: no Tunnels Other Applications: no

Excessive Bandwidth Use: yes Prone to Misuse: no

Used by Malware: no Widely Used: yes

Capable of File Transfer: no SaaS: yes

Has Known Vulnerabilities: yes

Options

Session Timeout (seconds): 30 [Customize...](#)

TCP Timeout (seconds): 3600 [Customize...](#)

TCP Half Closed (seconds): 120 [Customize...](#)

TCP Time Wait (seconds): 15 [Customize...](#)

App-ID Enabled: yes

Classification

Category: collaboration

Subcategory: internet-conferencing

Risk: 2

SaaS Characteristics

Certifications: HIPAA

Data Breaches: no

Restrictions: yes

Reliability: no

Service: no

Tags

Enterprise VoIP Web App

Predefined tags

Custom tags

Edit

- Palo Alto Networks assigns one or more tags to applications in App-ID:
 - Starting with PAN-OS 9.1
- Tags are updated as part of normal *Applications and Threats* content updates
- To use tags in Security policy:
 1. Use tags in application filters.
 2. Use application filters in Security policy.

Application filter filtering on Enterprise VoIP tag

	NAME	TAGS	Source		Destination		APPLICATION	SERVICE	ACTION
			ZONE	ADDRESS	ZONE	ADDRESS			
1	Permit-VOIP	Users_Net	Users_Net	any	Internet	any	Allow-VOIP	application-default	Allow

App-ID in Policy Rules Reduces the Attack Surface

Policies > Security

	NAME	TAGS	TYPE	Source			Destination		APPLICATION	SERVICE		
				ZONE	ADDRESS	USER	ZONE	ADDRESS				
1	Allow-Trusted	Users_Net	universal	Users_Net	any	any	Internet	any	Trusted-Applications	application-default	any	Allow
2	Allow-P2P	Users_Net	universal	Users_Net	any	IT-Grp	Internet	any	Custom-P2P-Apps	application-default	any	Allow
3	DNS	Users_Net	universal	Users_Net	any	any	Internet	any	dns	application-default	any	Allow

Application group

Application filter

Application

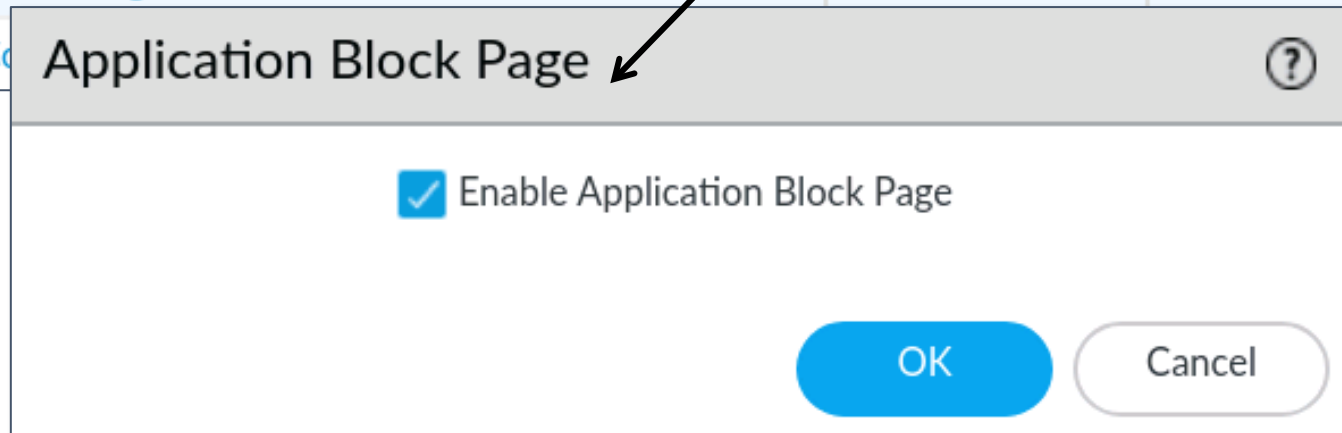
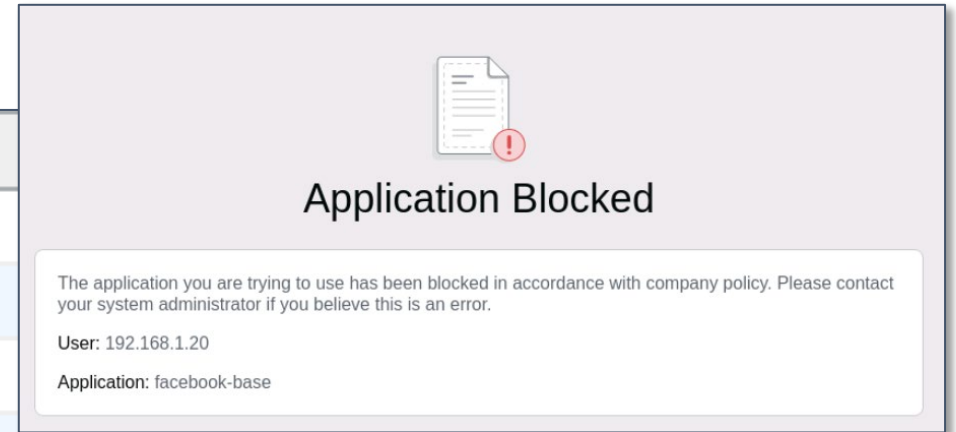
- Implement App-ID using a positive enforcement model:
 - Specify what to allow rather than what to block.
- Implement policy by application rather than by port:
 - Specify applications, application groups, application filters.

Application Block Page

For blocked web-based applications, a response page can be displayed in the user's browser.

Device > Response Pages

TYPE	ACTION
Antivirus / Anti-spyware Block Page	
Application Block Page	Enabled
Captive Portal Comfort Page	
Data Filtering Block Page	
File Blocking Co	



View Applications used in the Traffic Log

- App-ID labels traffic in logs and reports.
- The goal is to reduce the number of illegitimate or unknown applications allowed by the firewall.

Monitor > Logs > Traffic

		RECEIVE TIME	TYPE	FROM ZONE	TO ZONE	SOURCE	DESTINATION	TO PORT	APPLICATION	ACTION	RULE	SESSION END REASON	BYTES
		07/20 22:54:12	end	Users_Net	Internet	192.168.1.20	69.171.250.63	443	instagram-base	allow	Allow-Trusted	tcp-fin	17.0k
		07/20 22:54:12	end	Users_Net	Internet	192.168.1.20	69.171.250.174	443	instagram-base	allow	Allow-Trusted	tcp-fin	1.2M
		07/20 22:54:03	end	Users_Net	Internet	192.168.1.20	69.171.250.174	443	instagram-base	allow	Allow-Trusted	tcp-fin	8.2k
		07/20 22:53:49	end	Users_Net	Internet	192.168.1.20	184.31.10.237	443	twitter-base	allow	Allow-Trusted	tcp-fin	54.6k
		07/20 22:53:42	end	Users_Net	Internet	192.168.1.20	172.217.14.174	80	google-base	allow	Allow-Trusted	tcp-fin	912
		07/20 22:53:37	end	Users_Net	Internet	192.168.1.20	184.31.10.237	443	twitter-base	allow	Allow-Trusted	tcp-fin	7.1k
		07/20 22:53:27	end	Users_Net	Internet	192.168.1.20	184.31.10.237	443	twitter-base	allow	Allow-Trusted	tcp-fin	2.6k
		07/20 22:53:25	end	Users_Net	Internet	192.168.1.20	3.225.55.231	443	instagram-base	allow	Allow-Trusted	tcp-fin	12.9k
		07/20 22:53:18	end	Users_Net	Internet	192.168.1.20	192.229.210.163	443	twitter-base	allow	Allow-Trusted	tcp-fin	8.8k
		07/20 22:53:07	end	Users_Net	Internet	192.168.1.20	72.21.91.70	443	twitter-base	allow	Allow-Trusted	tcp-rst-from-client	3.4k

App-ID reduces the attack surface

App-ID concepts and operation

Configure App-ID related objects

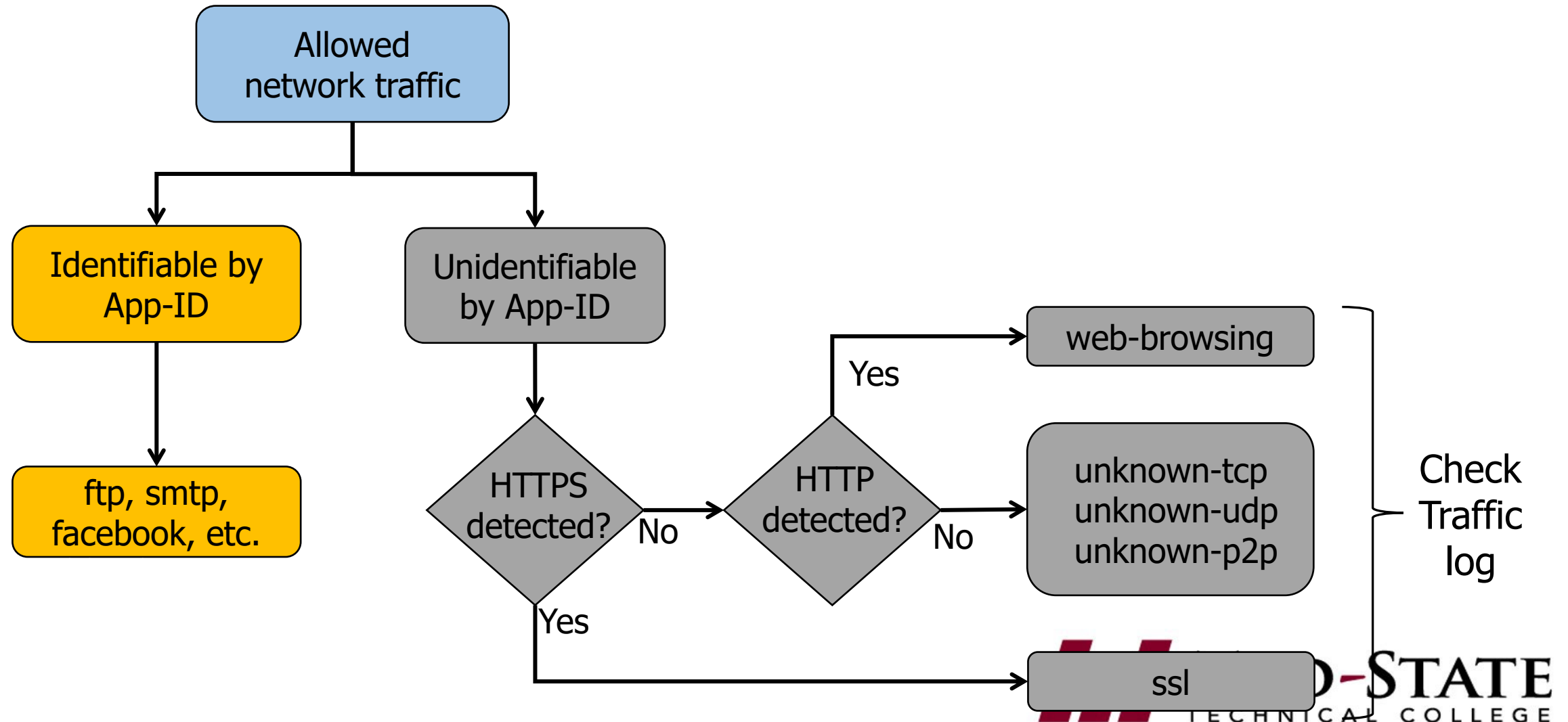
Unknown and encrypted application traffic



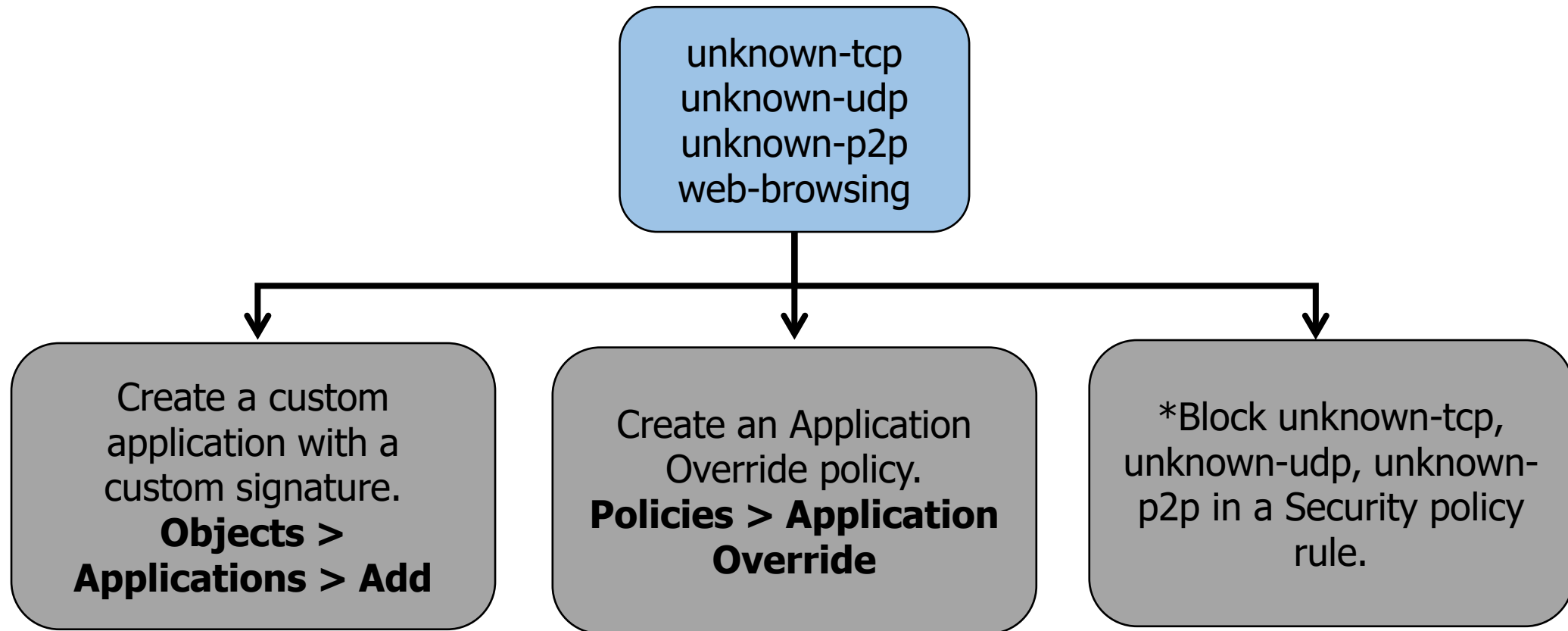
Migrating to an App-ID-based Security Policy

Updating App-ID

Differentiating Between Known and Unknown Applications



Control Unknown Applications



*Could block more traffic than intended

Control Applications on SSL-Secure Ports

- *Application-default* matches:
 - Cleartext applications on their **Standard Ports**
 - SSL-encrypted applications on their **Secure Ports**
- Supported applications are:
 - web-browsing
 - SMTP
 - FTP
 - LDAP
 - POP3
 - IMAP

For more information about this topic, see the module in this course on “Using decryption to Block Threats in Encrypted Traffic.”

Objects > Applications

Application
Name: web-browsing Standard Ports: tcp/80 Secure Ports: tcp/443 Depends on: Implicitly Uses: Deny Action: drop-reset Additional Information: Wikipedia Google Yahoo!

Control Applications on Non-Standard Ports

Policies > Security

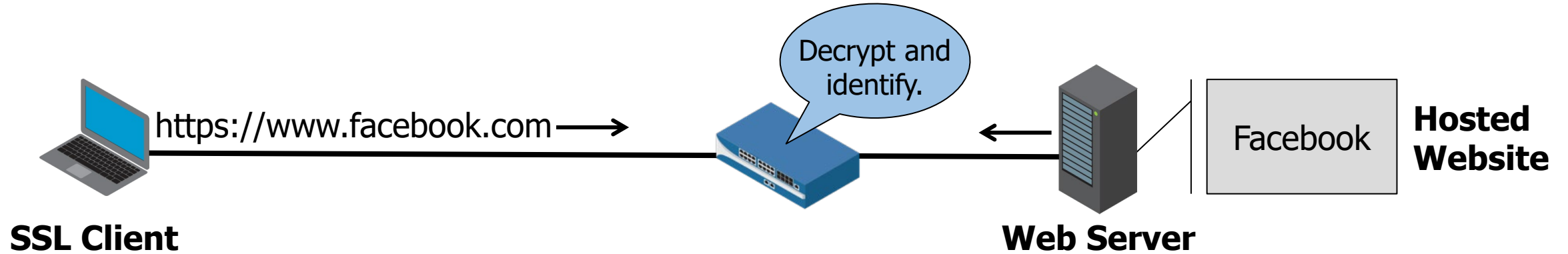
	NAME	TYPE	Source				Destination			APPLICATION	SERVICE	URL CATEGORY	ACTION
			ZONE	ADDRESS	USER	DEVICE	ZONE	ADDRESS	DEVICE				
1	Block-SSH-Evasion	universal	Users_Net	any	any	any	Internet	any	any	ssh ssh-tunnel	any	any	Deny
2	Allow-RDP	universal	Users_Net	any	any	any	Extranet	any	any	ms-rdp	application-default	any	Allow
3	Allow-MySQL	universal	Users_Net	any	any	any	Extranet	any	any	mysql	Service-MySQL-45451	any	Allow

- Malicious traffic often uses non-standard application ports.
- Rule 1 blocks SSH tunnels on the standard *or* non-standard ports.
- Rule 2 allows RDP traffic on *only* standard ports.
- Rule 3 allows mysql traffic only on a non-standard port.

Objects > Applications

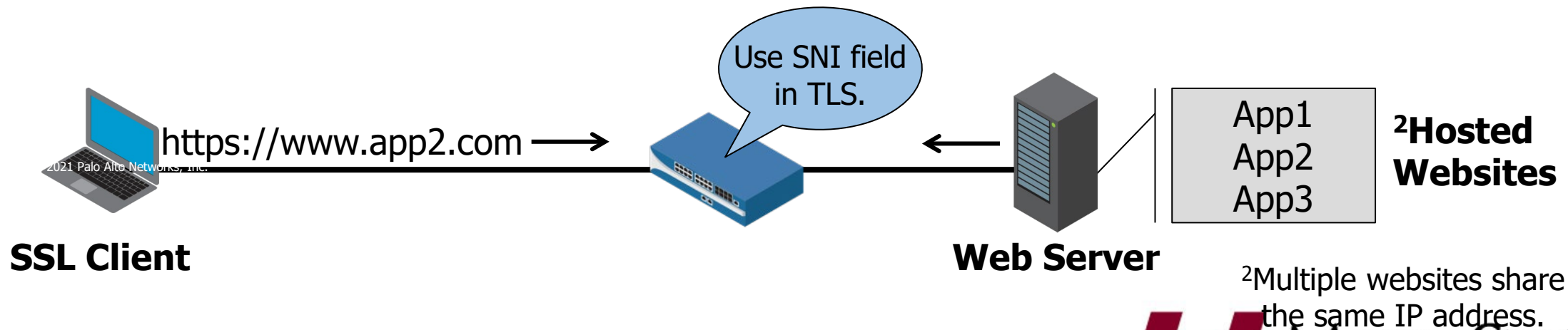
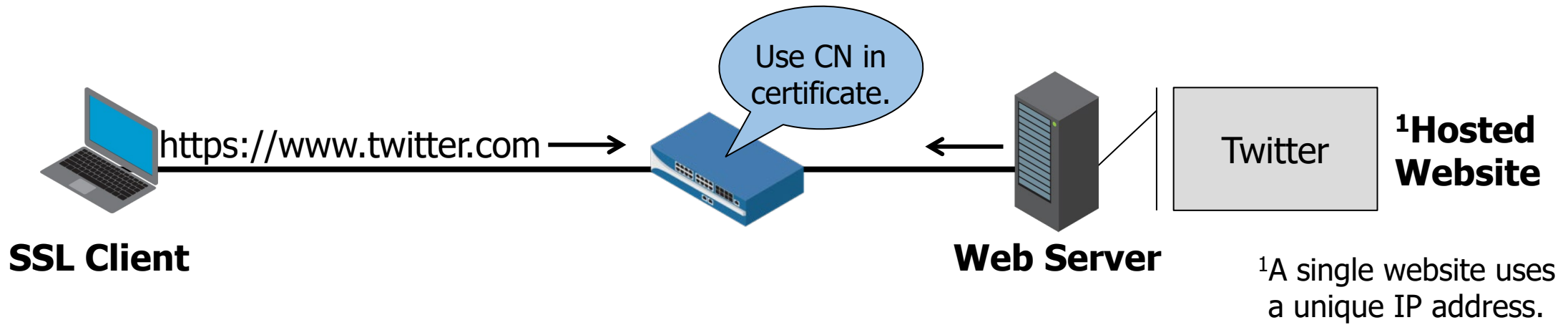
Application	
Name:	mysql
Standard Ports:	tcp/3306
Depends on:	
Implicitly Uses:	
Deny Action:	drop-reset
Additional Information:	Wikipedia Google Yahoo!

Identify Applications in Decrypted SSL Traffic



- SSL encrypts application-layer data.
- The firewall can identify and decrypt SSL traffic.
- App-ID identifies applications in decrypted SSL traffic:
 - Uses signatures, decoders, behavioral heuristics

Identify Applications in Encrypted SSL Traffic



App-ID reduces the attack surface

App-ID concepts and operation

Configure App-ID related objects

Unknown and encrypted application traffic

Migrating to an App-ID-based Security Policy



Updating App-ID

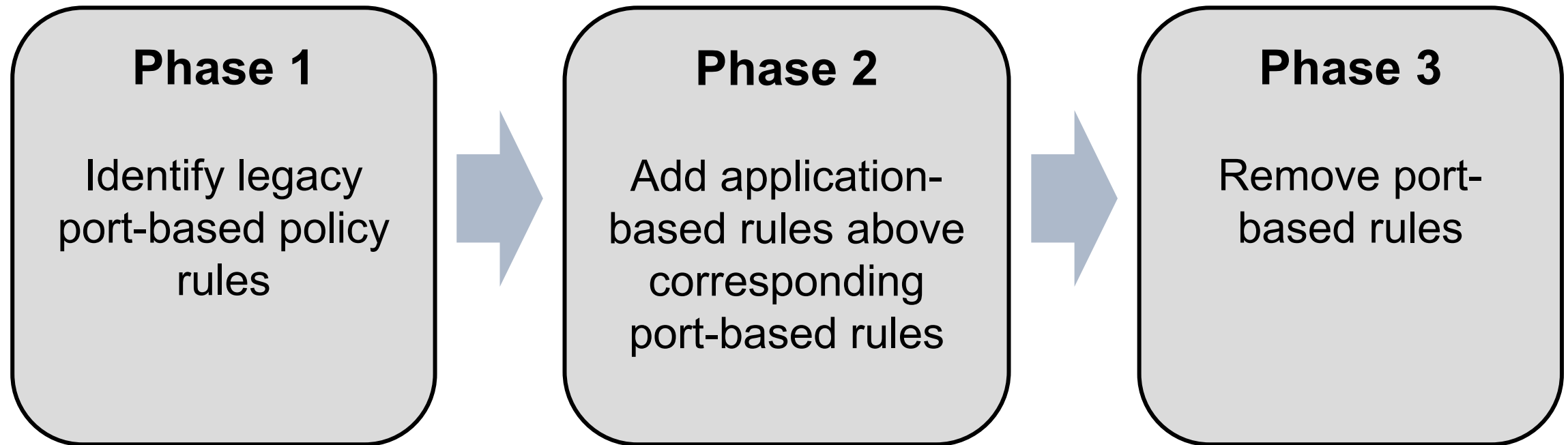
Policy Optimizer

- Migrate port-based rules to App-ID-based rules
- Help reduce attack surface and provide information about application usage
- Prevent evasive applications from running on non-standard ports
- Identify over-provisioned application-based rules

Policies > Security > Policy Optimizer > Rules Without App Controls

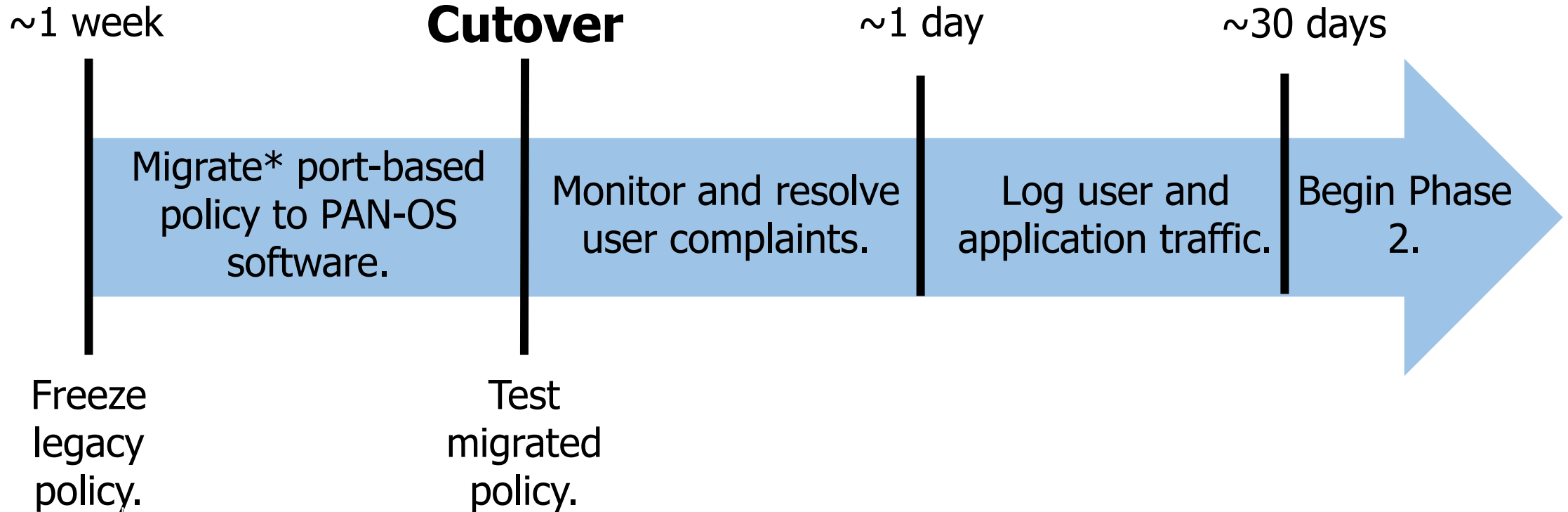
New App Viewer1+ Rules Without App Controls1 Unused Apps0 Log Forwarding for Security Services Rule Usage Unused in 30 days5 Unused in 90 days5 Unused5		Rules Without App Controls These rules require immediate attention to prevent unwanted and potentially dangerous applications from accessing your network! These port-based rules allow any application because they don't define specific applications. These rules may allow apps that you don't want and that present a security risk. Use Policy Optimizer to examine the applications that match these rules and to safely convert port-based rules to app-id-based rules that allow only the applications you want on your network. 1 item							
	NAME	SERVICE	TRAFFIC (BYTES, 30 DAYS)	App Usage				MODIFIED	CREATED
				APPS ALLOWED	APPS SEEN	DAYS WITH NO NEW APPS	COMPARE		
5	Allow_UserNet_to_Extranet	any	5.9M	any	3	0	Compare	2022-01-12 16:17:25	2022-01-12 16:17:25

Moving to Application-Based Policies



For a Tech Doc about this topic, log into Live and search for "Migrate Port-Based to App-ID Based Security Policy Rules."

Phase 1: Migrate Port-Based Rules



*Expedition tool provides some automation.

Phase 2: View Data of Port-Based Rules

Use **Rules Without App Controls** to discover port-based rules.

Policies > Security

	NAME	TAGS	Source		Destination		APPLICATION	SERVICE	ACTION
			ZONE	ADDRESS	ZONE	ADDRESS			
6	egress-outside-content-id	egress	Users_Net	any	Internet	any	any	application-default	Allow

Policies > Security > Policy Optimizer > Rules Without App Controls

Application "any" triggers **Rules Without App Controls** match.

Rules Without App Controls										
These rules require immediate attention to prevent unwanted and potentially dangerous applications from accessing your network! These port-based rules allow any application because they don't define specific applications. These rule and that present a security risk. Use Policy Optimizer to examine the applications that match these rules and to safely convert port-based rules to app-id-based rules that allow only the applications you want on your network.										
	NAME	SERVICE	TRAFFIC (BYTES, 30 DAYS)	App Usage				MODIFIED	CREATED	
				APPS ALLOWED	APPS SEEN	DAYS WITH NO NEW APPS	COMPARE			
4	egress-outside	application-d...	11.7G	any	10	5	Compare	2022-03-23 13:50:35	2022-03-17 17:40:15	
6	egress-outside-conte...	application-...	294.6M	any	7	0	Compare	2022-03-23 13:50:35	2022-03-18 14:59:30	
7	danger-simulated-tra...	application-...	13.6M	any	36	6	Compare	2022-03-17 17:40:15	2022-03-17 14:12:28	

Phase 2: Discover Applications Matching a Port-Based Rule

Policies > Security > Policy Optimizer > Rules Without App Controls

	NAME	SERVICE	TRAFFIC (BYTES, 30 DAYS)	App Usage				MODIFIED	CREATED
				APPS ALLOWED	APPS SEEN	DAYS WITH NO NEW APPS	COMPARE		
5	Allow_UserNet_to_Extranet	any	5.9M 	any	3	0	Compare	2022-01-12 16:17:25	2022-01-12 16:17:25

- Click **App Seen** number or **Compare** to view any applications that matched the port-based rule.
- The firewall displays a list of applications seen and identified by a rule.
- Use applications listed to create application-based rule(s).

Applications & Usage - Allow_UserNet_to_Extranet

Timeframe: Anytime

Apps on Rule

- ☒ Any
- ☐ APPLICATIONS

Apps Seen: 3

APPLICATIONS	SUBCATEGO...	RISK	FIRST SEEN	LAST SEEN	TRAFFIC (30 DAYS)
☐ dns	infrastructure	3	2022-01-12	2022-01-13	5.8M 
☐ ntp	infrastructure	2	2022-01-12	2022-01-13	60.7k 
☐ smtp-base	email	5	2022-01-13	2022-01-13	53.9k 

3 items

Anytime
Past 7 days
Past 15 days
Past 30 days

Browser + Add - Delete

Create Cloned Rule + Add to This Rule + Add to Existing Rule Match Usage

The last new app was discovered 0 days ago.

Four options to convert the policy

Phase 2: Clone a Port-Based Rule Using “Create Cloned Rule”

Option 1 of 4:

Applications & Usage - Allow_UserNet_to_Extranet

Timeframe: Anytime

Apps on Rule

- ☒ Any
- ☐ APPLICATIONS ^

Apps Seen: 3

APPLICATIONS	SUBCATEGORY	RISK	FIRST SEEN	LAST SEEN	TRAFFIC (30 DAYS)
☐ dns	infrastructure	3	2022-01-12	2022-01-13	5.8M
☐ ntp	infrastructure	2	2022-01-12	2022-01-13	60.7k
☒ smtp-base	email	5	2022-01-13	2022-01-13	50.0k

Buttons: Browse, Add, Delete, **Create Cloned Rule**, Add to This Rule, Add to Extranet

The last new app was discovered 0 days ago.

1. Select application(s).
2. Click **Create Cloned Rule**.
3. Name new rule.

- Clones port-based rule to new application-based rule
- Safest method when many applications permitted by a rule
- Lists and prompts for required application dependencies

Create Cloned Rule

Name:

Applications

☒ Add container app ☐ Add specific apps seen

APPLICATION	LAST SEEN
☒ smtp	
☒ smtp-base	2022-01-13
☒ smtp-starttls	--

Name the new rule.

Result of “Create Cloned Rule”

Applications & Usage - Users_to_Extranet

Timeframe Anytime

Apps on Rule Apps Seen 3

Any

APPLICATIONS

dns

web-browsing

syslog

infrastructure

internet-utility

management

3

4

2

2020-07-21

2020-07-21

2020-07-21

2020-07-21

2020-07-21

2020-07-21

506.2k

55.0k

3.9k

3 items

Browse

Add

Delete

Create Cloned Rule

Add to This Rule

Add to Existing Rule

Match Usage

The last new app was discovered 0 days ago.

The **ftp** application is removed from the port-based rule **Apps Seen** list and placed in a new rule.

Must manually configure as **application-default**

Policies > Security

	NAME	TAGS	Source		Destination		APPLICATION	SERVICE	ACTION
			ZONE	ADDRESS	ZONE	ADDRESS			
1	Allow-FTP	Users_Net	Users_Net	any	Extranet	any	ftp	service-ftp	Allow
								service-http	
2	Users_to_Extranet	Users_Net	Users_Net	any	Extranet	any	any	service-ftp	Allow
								service-http	

Phase 2: Replace a Port-Based Rule Using “Add to This Rule”

Option 2 of 4:

Applications & Usage - Users_to_Extranet

Timeframe: Anytime

Apps on Rule: Apps Seen: 4

4 items → ×

☒ Any ☐ AP

1. Select application(s).

2. Click **Add to This Rule**.

	RISK	FIRST SEEN	LAST SEEN	TRAFFIC (30 DAYS)
☐		2020-07-21	2020-07-21	506.2k
☐		2020-07-21	2020-07-21	55.0k
☐		2020-07-21	2020-07-21	3.9k
☒		2020-07-21	2020-07-21	3.1k

syslog management 2

ftp file-sharing 5

⏮ Browse ⏹ Add ⏻ Delete ⚙ Create Cloned Rule ➕ Add to This Rule ➕ Add to Existing Rule

- Firewall *replaces* port-based rule by moving *selected* applications to a new rule.
- Riskier method because some required applications could be inadvertently missed.

Add to This Rule

Applications

☒ Add container app ☐ Add specific apps seen

☒	APPLICATION ^	LAST SEEN
☒	ftp	2020-07-21

Result of “Add to This Rule”

Policies > Security

	NAME	TAGS	TYPE	Source			Destination				URL CATEGORY	ACTION
				ZONE	ADDRESS	USER	ZONE	ADDRESS	APPLICATION	SERVICE		
1	Users_to_Extranet	Users_Net	universal	Users_Net	any	any	Extranet	any	ftp	service-ftp service-http	any	Allow

Must manually configure
as **application-default**

New application-based rule
replaces port-based rule.

Phase 2: Replace a Port-Based Rule Using “Add to Existing Rule”

Option 3 of 4:

The screenshot displays the Palo Alto Networks management interface. The main window is titled "Applications & Usage - Users_to_Extranet". It features a "Timeframe" dropdown set to "Anytime" and a table of applications. The table has columns for "Apps on Rule", "Apps Seen", "APPLICATIONS", "SUBCATEGORY", "RISK", and "FIRST SEEN". The "ftp" application is selected. An overlay dialog titled "Add Apps to Existing Rule" is open, showing a list of applications to add. A callout box with two steps is present: "1. Select application(s). 2. Click **Add to Existing Rule**." Arrows point from the "Add to Existing Rule" button in the main interface and the second step of the callout to the "Add to Existing Rule" button in the dialog.

Applications & Usage - Users_to_Extranet

Timeframe: Anytime

Apps on Rule: 4

Apps Seen: 4

1. Select application(s).
2. Click **Add to Existing Rule**.

Add Apps to Existing Rule

Name: [Dropdown]

Application List:

- 1 - Users_to_Extranet
- 2 - Users_Net-Apps
- 3 - Users_to_Internet
- 4 - Extranet_to_Internet
- 5 - Allow-PANW-Apps

Buttons: [Add] [Add specific apps seen]

LAST SEEN: 2020-07-21

Buttons: Browse, Add, Delete, Create Cloned Rule, Add to This Rule, **Add to Existing Rule**, Match Usage

Result of “Add to Existing Rule”

Policies > Security

	NAME	TAGS	TYPE	Source			Destination		APPLICATION	SERVICE	URL CATEGORY	ACTION
				ZONE	ADDRESS	USER	ZONE	ADDRESS				
1	Users_to_Extranet	Users_Net	universal	Users_Net	any	any	Extranet	any	any	service-ftp service-http	any	Allow
2	Users_Net-Apps	Users_Net	universal	Users_Net	any	any	Extranet	any	dns ftp ssh ssl	application-default	any	Allow

No change to port-based rule

Existing internal-apps rule *modified* to include ftp application

Existing rule was already configured for **application-default**.

Phase 2: Replace a Port-Based Rule Using “Match Usage”

Option 4 of 4:

Applications & Usage - Users_to_Extranet

Timeframe: Anytime

Apps on Rule: ☐ Any

Apps Seen: 4

APPLICATIONS	SUBCATEGO...	RISK	FIRST SEEN	LAST SEEN	TRAFFIC (30 DAYS)
☐ dns	infrastructure	3	2020-07-21	2020-07-21	506.2k
☐ web-browsing			2020-07-21	2020-07-21	55.0k
☐ syslog			2020-07-21	2020-07-21	3.9k
☐ ftp			2020-07-21	2020-07-21	3.1k

Annotations:

- All applications are added to the left-side **Apps on Rule** column.
- Click **Match Usage**.

Buttons: Browse, Add, Delete, Create Cloned Rule, Add to This Rule, Add to Existing Rule, Match Usage

The last new app was discovered 0 days ago.

- Use only when the rule matches a small number of legitimate applications.
- Copies *all* applications under **Apps Seen** to **Apps on Rule**.
- Firewall *replaces* port-based rule with application-based rule.

Result of “Match Usage”

Policies > Security

	NAME	TAGS	TYPE	Source			Destination		APPLICATION	SERVICE	URL CATEGORY	ACTION
				ZONE	ADDRESS	USER	ZONE	ADDRESS				
1	Users_to_Extranet	Users_Net	universal	Users_Net	any	any	Extranet	any	dns ftp syslog web-browsing	service-ftp service-http	any	Allow

Must manually configure
as **application-default**

New application-based rule
replaces port-based rule.

Prioritize Port-Based Rules to Convert

Rules Without App Controls

Prioritize rules passing more data.

Prioritize rules with more applications.

Prioritize rules that are more stable.

	NAME	SERVICE	TRAFFIC (BYTES, 30 DAYS)	App Usage			COMPARE	MODIFIED	CREATED
				APPS ALLOWED	APPS SEEN	DAYS WITH NO NEW APPS			
4	egress-outside	application-d...	11.7G	any	10	5	Compare	2022-03-23 13:50:35	2022-03-17 17:40:15
6	egress-outside-conte...	application-...	294.6M	any	7	0	Compare	2022-03-23 13:50:35	2022-03-18 14:59:30
7	danger-simulated-tra...	application-...	13.6M	any	36	6	Compare	2022-03-17 17:40:15	2022-03-17 14:12:28

Security

Policy Optimizer

Rule Usage

These rules re... selected time... you need to a... meetings, con...

Timeframe All time Usage Unused Exclude rules reset during the last 90 days

Prioritize rules that match more sessions.

	NAME	Rule Usage				MODIFIED	CREATED
		HIT COUNT	LAST HIT	FIRST HIT	RESET DATE		
5	egress-outside-app-id	0	-	-	-	2022-03-18 14:59:30	2022-03-18 14:59:30
7	internal-dmz-ftp	0	-	-	-	2022-03-18 14:59:30	2022-03-18 14:59:30

Phase 3: Review Port-Based Rules

- After 60 days, review the **Hit Count** columns in the Security policy.
- Look for port-based rules with zero hits.

Policies > Security

	NAME	TAGS	TYPE	Source			Destination		APPLICATION	SERVICE	URL CATEGORY	ACTION	PROFILE	OPTIONS	HIT COUNT
				ZONE	ADDRESS	USER	ZONE	ADDRESS							
1	Users_Net-Apps	Users_Net	universal	Users_Net	any	any	Extranet	any	dns ftp ssh ssl	application-default	any	Allow	none		1250
2	Users_to_Extranet	Users_Net	universal	Users_Net	any	any	Extranet	any	any	service-ftp service-http			none		0

Reset

All rules
Selected rules

+ Add - Delete Clone Override Revert Enable Disable Move PDF/CSV Highlight Unused Rules View Rulebase as Groups Reset Rule Hit Counter Group Test Policy Match

Policies > Security

	NAME	TAGS	TYPE	Source			Destination		APPLICATION	SERVICE	URL CATEGORY	ACTION	PROFILE	OPTIONS	HIT COUNT
				ZONE	ADDRESS	USER	ZONE	ADDRESS							
1	Users_Net-Apps	Users_Net	universal	Users_Net	any	any	Extranet	any	dns ftp ssh ssl	application-default	any	Allow	none		1250
2	Users_to_Extranet	Users_Net	universal	Users_Net	any	any	Extranet	any	any	service-ftp service-http	any	Allow	none		0

+

Add

-

Delete

+

Clone

+

Override

+

Revert

+

Enable

+

Disable

+

Move

+

PDF/CSV

+

Highlight Unused Rules

+

View Rulebase as Groups

+

Reset Rule Hit Counter

+

Group

+

Test Policy Match

- Disable port-based rules that have not matched to any new traffic.
- Disabled rules are rendered in gray italic font.
- Tag rules that must be removed later (optional).

Phase 3: Remove Port-Based Rules

- After 90 days, delete port-based rules that have not matched to any new traffic.
- The goals:
 - At least 80% application-based rules
 - No inbound or outbound *unknown* applications (internal is acceptable)

Policies > Security

	NAME	TAGS	TYPE	Source			Destination		APPLICATION	SERVICE	URL CATEGORY	ACTION	PROFILE	OPTIONS	HIT COUNT
				ZONE	ADDRESS	USER	ZONE	ADDRESS							
1	Users_Net-Apps	Users_Net	universal	Users_Net	any	any	Extranet	any	dns ftp ssh ssl	application-default	any	Allow	none		1250
2	Users_to_Extranet	Users_Net	universal	Users_Net	any	any	Extranet	any	any	service-ftp service-http	any	Allow	none		0

+ Add ⊖ Delete 🔗 Clone 🔄 Override ↶ Revert ✅ Enable ❌ Disable 📁 Move 📄 PDF/CSV ☐ Highlight Unused Rules ☐ View Rulebase as Groups Reset Rule Hit Counter Group Test Policy Match



	NAME	TAGS	TYPE	Source			Destination		APPLICATION	SERVICE	URL CATEGORY	ACTION	PROFILE	OPTIONS	HIT COUNT
				ZONE	ADDRESS	USER	ZONE	ADDRESS							
1	Users_Net-Apps	Users_Net	universal	Users_Net	any	any	Extranet	any	dns ftp ssh ssl	application-default	any	Allow	none		1250
2	Users_to_Internet	Users_Net	universal	Users_Net	any	any	Extranet	any	any	application-default	any	Allow	none		896291
3	Extranet_to_Internet	none	universal	Extranet	any	any	Internet	any	any	application-default	any	Allow	none		0

+ Add ⊖ Delete 🔗 Clone 🔄 Override ↶ Revert ✅ Enable ❌ Disable 📁 Move 📄 PDF/CSV ☐ Highlight Unused Rules ☐ View Rulebase as Groups Reset Rule Hit Counter Group Test Policy Match

For a Tech Doc about this topic, log into Live and search for "Best Practices for Migrating to Application-Based Policy."

App-ID reduces the attack surface

App-ID concepts and operation

Configure App-ID related objects

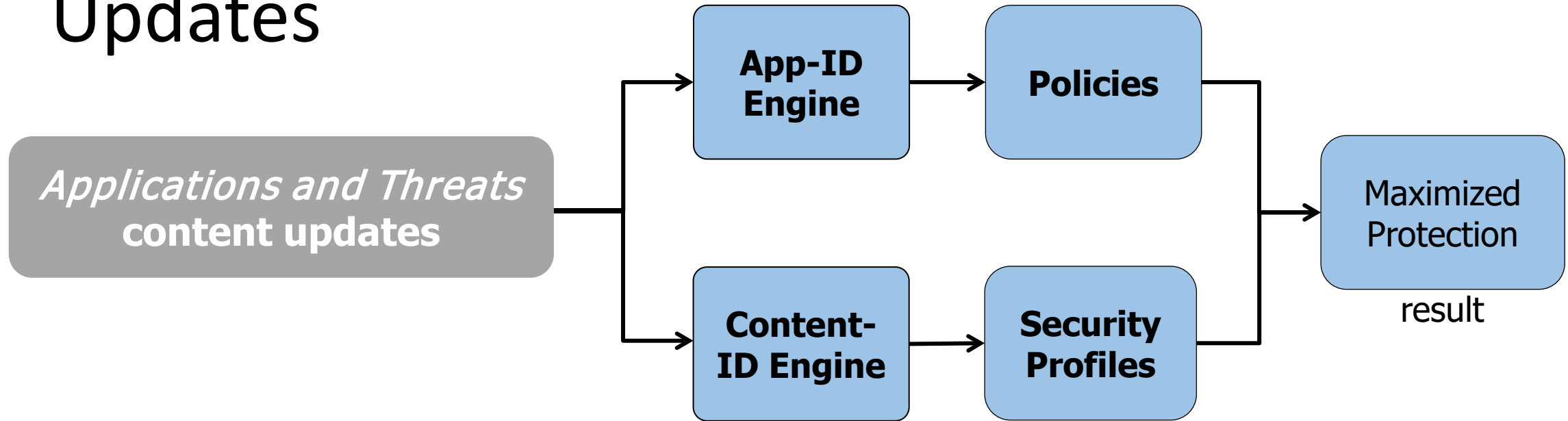
Unknown and encrypted application traffic

Migrating to an App-ID-based Security Policy

Updating App-ID



App-ID and Content-ID Depend on Content Updates



- To maintain optimal protection, schedule regular content updates.
- If App-ID cannot identify the traffic, then Content-ID cannot inspect the traffic.

For a Tech Doc about this topic, log into Live and search for "Deploy Application and Threats Content Updates"

Schedule Download and Install

Click to schedule updates.

Device > Dynamic Updates

Applications and Threats

Last checked: 2022/03/04 21:52:04 UTC

Schedule: Every hour at 52 minutes past the hour (Download and Install)

8470-6982	panupv2-all-contents-8470-6982	Apps, Threats	2021/10/09 17:19:49 UTC	Revert	Release Notes
8524-7228	panupv2-all-contents-8524-7228	Apps, Threats	2022/02/07 23:11:39 UTC	Download	Release Notes
8525-7229					Release Notes
8526-7230					Release Notes

Applications and Threats Update Schedule

Recurrence

Hourly

Minutes Past Hour

52

Action

download-and-install

☐ Disable new apps in content update

Old (hours)

[1 - 336]

A content update must be at least this many hours old for the action to be taken.

Allow Extra Time to Review New App-IDs

Set the amount of time the firewall waits before installing content updates that contain new App-IDs. You can use this wait period to assess and adjust your security policy based on the new App-IDs.

New App-ID Threshold (hours)

[1 - 336]

None (Manual)

Every 30 Minutes

Hourly

Daily

Weekly

None

download-only

download-and-install

If selected, new application signatures are disabled.

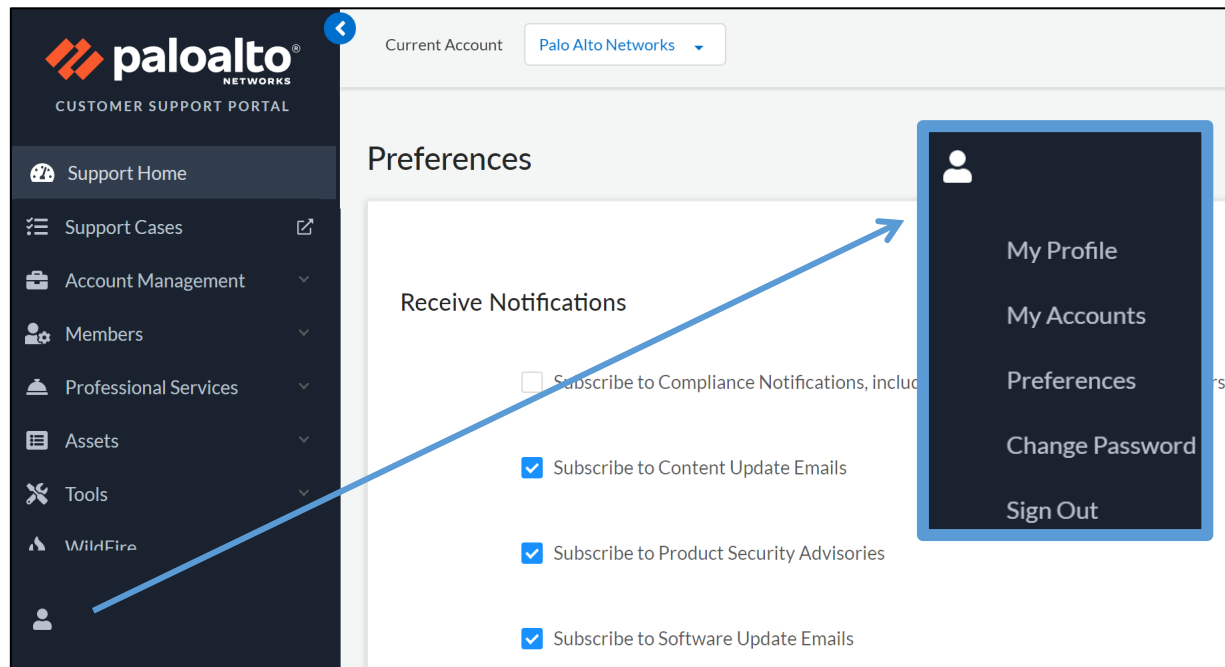


Review Content Update Release Notes

Device > Dynamic Updates

VERSION ▾	FILE NAME	FEATURES	RELEASE DATE	CURRENTLY INSTALLED	ACTION	DOCUMENTATION
Applications and Threats						
Last checked: 2022/03/04 21:52:04 UTC Schedule: Every hour at 52 minutes past the hour (Download and Install)						
8537-7273	panupv2-all-contents-8537-7273	Apps, Threats	2022/03/03 21:02:04 UTC		Install Review Policies Review Apps	Release Notes

<https://support.paloaltonetworks.com>



- Content Release Notes describe Security policy impacts.
- Get Content Release Notes from:
 - Firewall
 - Customer Support email

For a Tech Doc about this topic, log into Live and search for "Best Practices for Applications and Threat Content Updates"

Review New and Updated Application Details

Device > Dynamic Updates

VERSION	FILE NAME	FEATURES	RELEASE DATE	CURRENTLY INSTALLED	ACTION	DOCUMENTATION
Applications and Threats						
Last checked: 2022/03/04 21:52:04 UTC Schedule: Every hour at 52 minutes past the hour (Download and Install)						
8537-7273	panupv2-all-contents-8537-7273	Apps, Threats	2022/03/03 21:02:04 UTC		Install Review Policies Review Apps	Release Notes

New and Modified Applications since last installed content

159 items

New Apps

Content Version: 8476

abbott-serial

free-conference-call

free-conference-call-base

free-conference-call-uploading

hart-ip

hill-rom-nurse-call

silex-webserver

stryker-axeda-gateway

visme

visme-base

visme-uploading

Content Version: 8489

confluent

lenel-isc

Content Version: 8536-7270

Name: abbot-serial

Standard Ports: tcp/3004,3002

Depends on:

Implicitly Uses:

Previously Identified As: unknown-tcp

Deny Action: drop-reset

Additional Information: [Abbott](#) [Google](#) [Yahoo!](#)

Characteristics

Evasive: no

Excessive Bandwidth Use: no

Used by Malware: no

Capable of File Transfer: no

Has Known Vulnerabilities: no

Tunnels Other Applications: no

Prone to Misuse: no

Widely Used: no

Classification

Category: business-systems

Subcategory: medical

Risk: 1

Description:

Abbott devices exchange various information such as device settings, patient info and test results with a centralized server. This information also contains the communicating devices serial number. This App-ID covers Abbott device traffic in which the device serial number is present.

Options

TCP Timeout (seconds): 3600

TCP Half Closed (seconds): 120

TCP Time Wait (seconds): 15

App-ID Enabled: yes

Review Policies

Close

Review new and modified application information.

Review Policies

For a Tech Doc about this topic, log into Live and search for "See How New and Modified App-IDs Impact Your Security Policy."

Device > Dynamic Updates

VERSION ▾	FILE NAME	FEATURES	RELEASE DATE	CURRENTLY INSTALLED	ACTION	DOCUMENTATION
▾ Applications and Threats Last checked: 2022/03/04 21:52:04 UTC Schedule: Every hour at 52 minutes past the hour (Download and Install)						
8537-7273	panupv2-all-contents-8537-7273	Apps, Threats	2022/03/03 21:02:04 UTC		Install Review Policies Review Apps	Release Notes

Policy review based on candidate configuration

Content Version: 8296-6207 ▾ Rulebase: Security ▾ Virtual System: vsys1 ▾ Type: New Applications ▾ Application: crowdstrike ▾ ☐ Include rules with

	NAME	TAGS	TYPE	ZONE	ADDRESS	USER	DEVICE	ZONE	ADDRESS	DEVICE	APPLICATION	SERVICE	ACTION	PROFILE
☐	Users_Net-Apps	Users_Net	universal	Users_Net	any	any	any	Extranet	any	any	dns ftp ssh ssl web-browsing	application-default	Allow	none

Security
QoS
Policy Based Forwarding
SD-WAN

New Applications
Modified Applications

Enabled
abb-rp570
abbott-poca
amazon-prime-video
appletvplus
azure-govt-key-vault
beyond-trust-remote-suppo
cisco-viptela-ipsec-esp
classdojo
control-m
crowdstrike
disco

Add app to selected policies Remove app from selected policies Close

Review the policy impact of new or modified applications.

Module Summary

Now that you have completed this module, you should be able to:

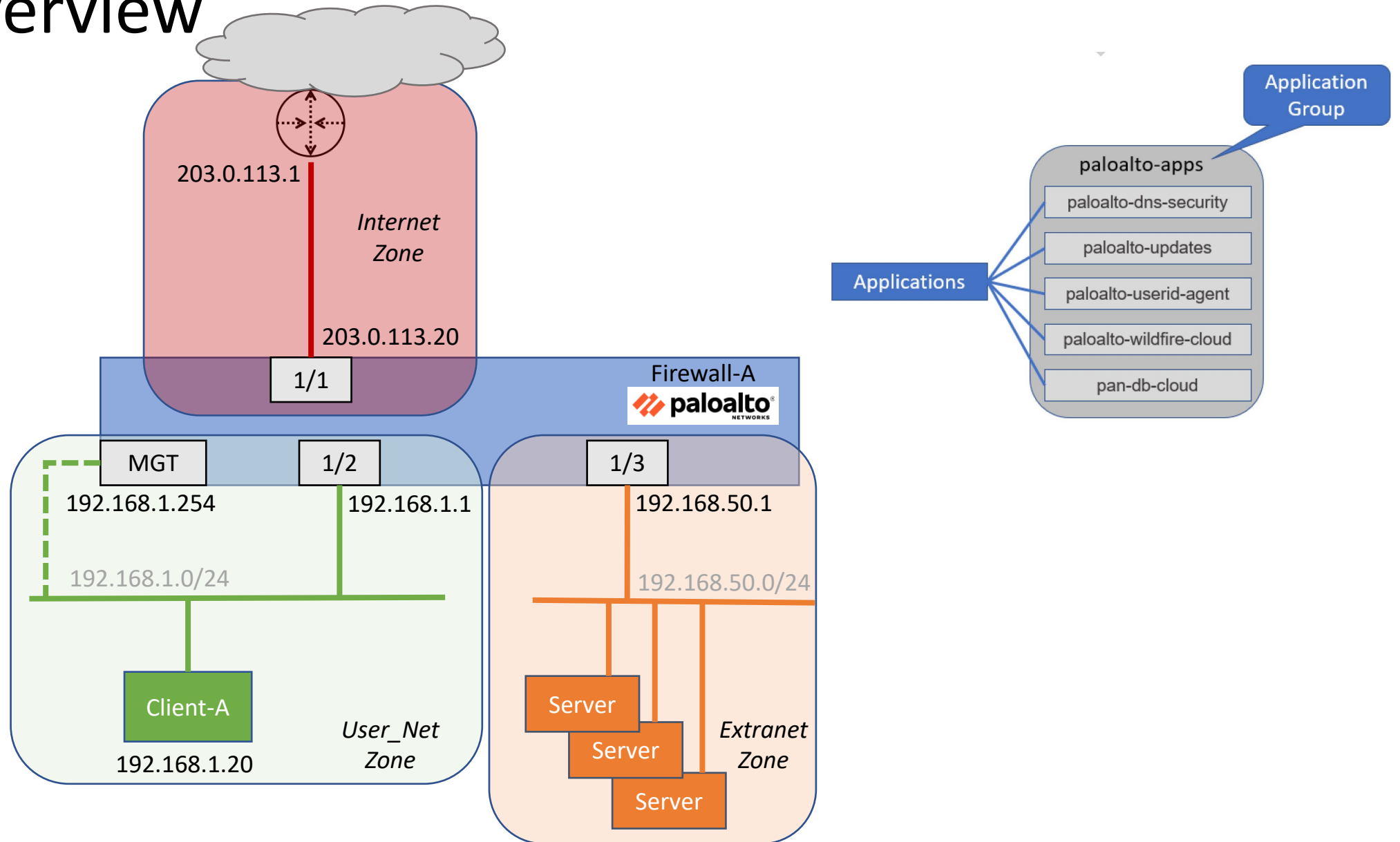
- Identify how App-ID reduces the attack surface
- Describe App-ID concepts and operation
- Configure App-ID-based policy rules
- Update App-ID application database



Questions



Lab 7 Overview



Lab 7: Controlling Application Usage with App-ID

- Load a baseline configuration
- Generate application traffic
- Configure an application group
- Configure a Security policy to allow update traffic
- Test the Allow-PANW-Apps Security policy rule
- Examine the tasks list to see shadowed message
- Modify the Security policy to function properly
- Test the modified Security policy rule

End Content

Unit 8 Homework

- Always check Blackboard for assignments and when they are due.
- Unit 8 Lab: Netlabs Lab 7 – Creating and managing NAT policy rules
- Unit 8 PAN App Control Video Worksheet
- Unit 8 Quiz